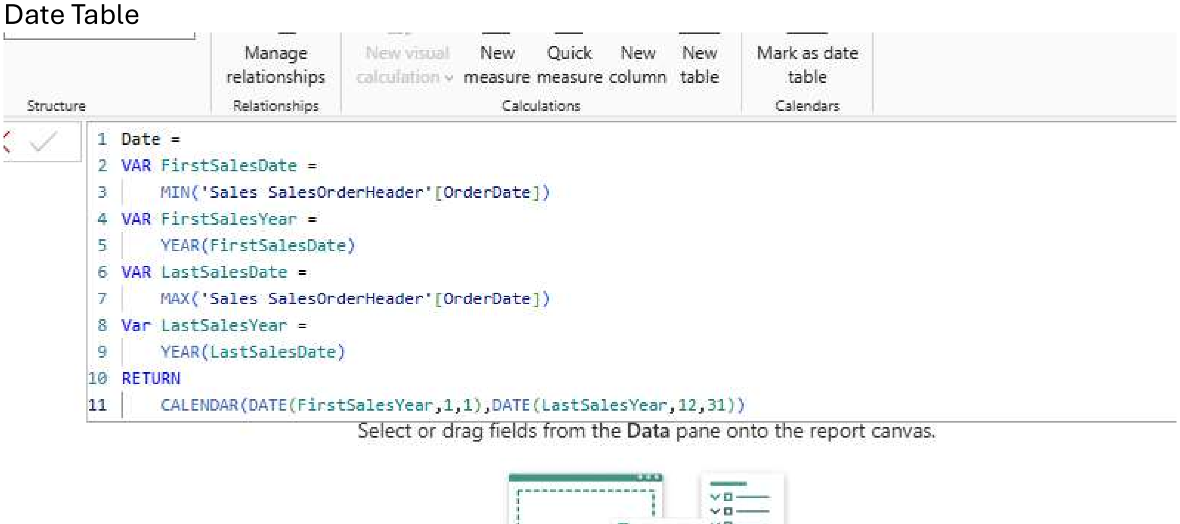
DAX Mastery Instructions

This document represents a comprehensive DAX portfolio developed to demonstrate practical expertise in data modeling and analytical problem-solving using Power BI. The work is based on two datasets:
AdventureDB (XLSX), used for sales, product, customer, and time intelligence analysis, and the **EMP-Sales DATASET**, used for hierarchical and organizational reporting scenarios.

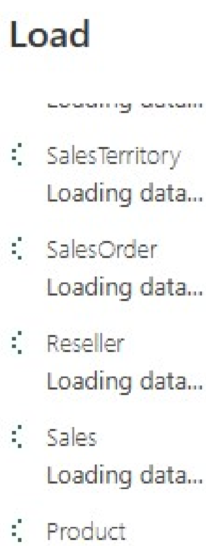
All models and calculations are developed in Power BI, with source files organized under the **"/Source Data/PowerBI DAX Source"** folder.

The portfolio showcases a wide range of real-world DAX patterns—including calculated columns, measures, evaluation contexts, advanced filtering, virtual tables, ranking, and time intelligence—demonstrating the ability to translate business requirements into optimized, scalable, and production-ready BI solutions.

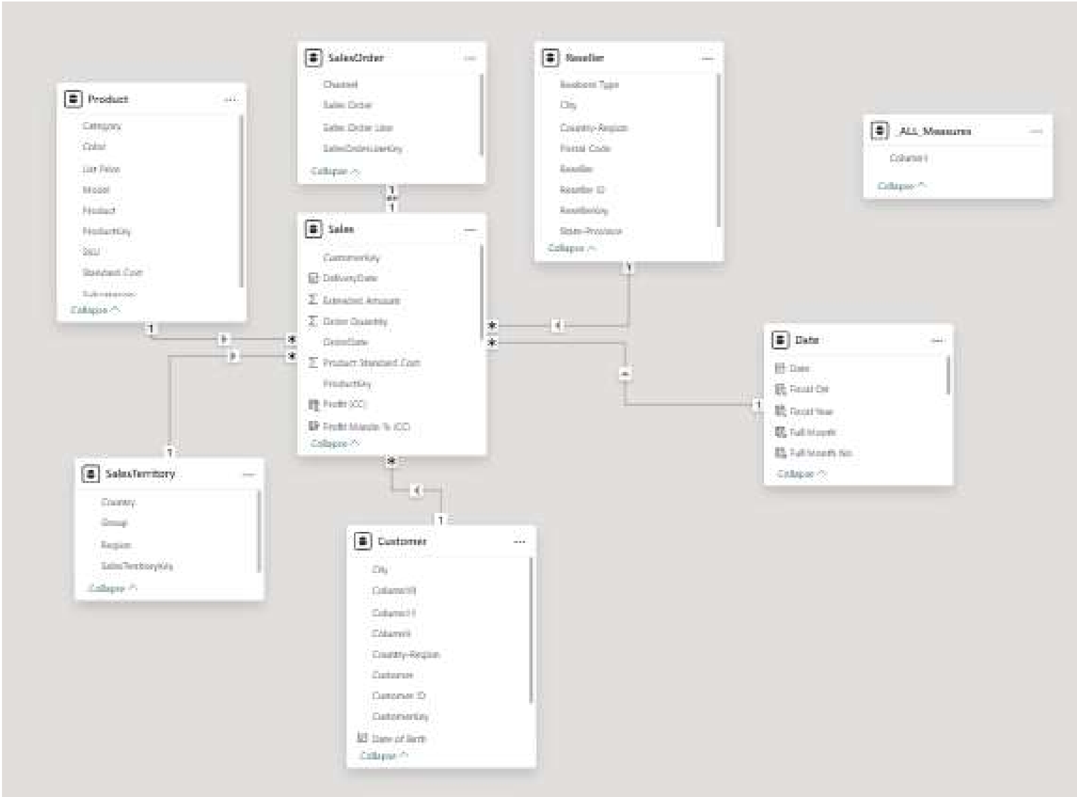
Date Table




```
Date =
VAR FirstSalesDate =
    MIN('Sales SalesOrderHeader'[OrderDate])
VAR FirstSalesYear =
    YEAR(FirstSalesDate)
VAR LastSalesDate =
    MAX('Sales SalesOrderHeader'[OrderDate])
Var LastSalesYear =
    YEAR(LastSalesDate)
RETURN
    CALENDAR(DATE(FirstSalesYear,1,1),DATE(LastSalesYear,12,31))
```



Model View:



Calculated Column:

Profit (CC) = Sales[Sales Amount]-Sales[Total Product Cost]

Profit Margin (CC) = Sales[Profit (CC)]/Sales[Sales Amount]

Category	Sum of Sales Amount	Sum of Profit (CC)	Sum of Profit Margin % (CC)
Components	**11,799,074.18**	**$1,032,957.07**	**272340.82%**
Clothing	**2,117,613.52**	**$368,846.60**	**502739.41%**
Vests	259,488.51	$99,466.08	66968.08%
Tights	201,832.98	$60,715.09	29895.19%
Socks	29,745.21	$12,164.99	68068.82%
Shorts	413,522.68	$155,941.17	118307.97%
Jerseys	752,257.82	($92,911.64)	3233.19%
Gloves	242,796.58	$83,411.82	159880.87%
Caps	51,230.10	($1,198.49)	32878.30%
Bib-Shorts	166,739.64	$51,257.58	23506.97%
Bikes	**94,620,526.59**	**$10,515,090.51**	**463560.54%**
Touring Bikes	14,296,290.29	$217,280.70	14826.76%
Road Bikes	43,878,791.78	$4,364,897.96	255029.35%
Mountain Bikes	36,445,444.52	$5,932,911.85	193704.43%
Accessories	**1,272,059.71**	**$634,391.75**	**2434803.49%**
Tires and Tubes	246,454.42	$154,016.23	1090366.11%
Pumps	13,514.71	$4,196.75	8323.55%
Locks	16,225.22	$5,025.51	8066.47%
Hydration Packs	105,826.62	$49,040.07	58520.29%
Helmets	484,049.97	$226,383.23	495361.77%
Fenders	46,619.58	$29,184.96	132779.62%
Cleaners	18,407.03	$8,541.62	72367.53%
Bottles and Cages	64,274.95	$38,214.28	515653.01%
Bike Stands	39,591.00	$24,782.97	15586.77%
Bike Racks	237,096.21	$95,006.13	37778.38%
Total	**109,809,274.00**	**$12,551,285.93**	**3673444.26%**

Notice that none of the aggregation available suits the requirement of percentage $\sum Profit$ calculation because

$$\sum\left(\frac{Profit}{Sales\ Amount}\right) \neq \frac{\sum Profit}{\sum Sales\ Amount}.$$

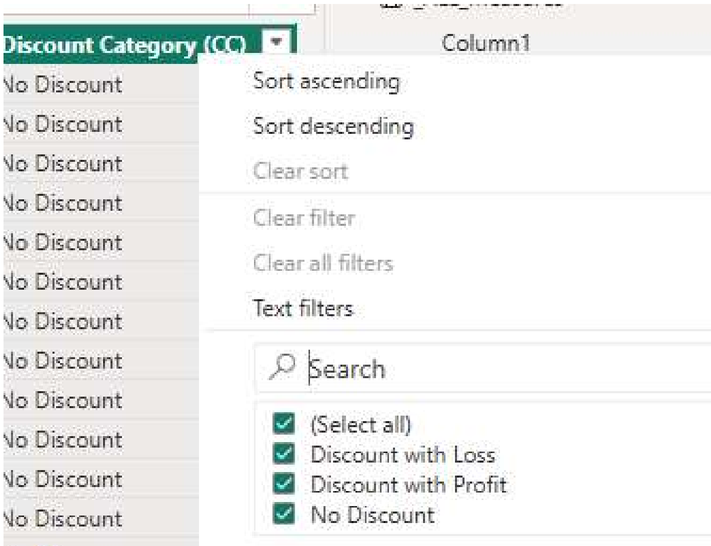
Calculated Measure:
Profit (CM) = SUM(Sales[Sales Amount]) - sum(Sales[Total Product Cost])
Profit Margin % (CM) = [Profit (CM)]/SUM(Sales[Sales Amount])

Category	Sum of Sales Amount	Sum of Profit (CC)	Sum of Profit Margin % (CC)	Profit (CM)	Profit Margin % (CM)
Components	**11,799,074.18**	**$1,032,957.07**	**272340.82%**	**1,032,957.07**	**8.75%**
Clothing	**2,117,613.52**	**$368,846.60**	**502739.41%**	**368,846.60**	**17.42%**
Vests	259,488.51	$99,466.08	66968.08%	99,466.08	38.33%
Tights	201,832.98	$60,715.09	29895.19%	60,715.09	30.08%
Socks	29,745.21	$12,164.99	68068.82%	12,164.99	40.90%
Shorts	413,522.68	$155,941.17	118307.97%	155,941.17	37.71%
Jerseys	752,257.82	($92,911.64)	3233.19%	-92,911.64	-12.35%
Gloves	242,796.58	$83,411.82	159880.87%	83,411.82	34.35%
Caps	51,230.10	($1,198.49)	32878.30%	-1,198.49	-2.34%
Bib-Shorts	166,739.64	$51,257.58	23506.97%	51,257.58	30.74%
Bikes	**94,620,526.59**	**$10,515,090.51**	**463560.54%**	**10,515,090.51**	**11.11%**
Touring Bikes	14,296,290.29	$217,280.70	14826.76%	217,280.70	1.52%
Road Bikes	43,878,791.78	$4,364,897.96	255029.35%	4,364,897.96	9.95%
Mountain Bikes	36,445,444.52	$5,932,911.85	193704.43%	5,932,911.85	16.28%
Accessories	**1,272,059.71**	**$634,391.75**	**2434803.49%**	**634,391.75**	**49.87%**
Tires and Tubes	246,454.42	$154,016.23	1090366.11%	154,016.23	62.49%
Pumps	13,514.71	$4,196.75	8323.55%	4,196.75	31.05%
Locks	16,225.22	$5,025.51	8066.47%	5,025.51	30.97%
Hydration Packs	105,826.62	$49,040.07	58520.29%	49,040.07	46.34%
Helmets	484,049.97	$226,383.23	495361.77%	226,383.23	46.77%
Fenders	46,619.58	$29,184.96	132779.62%	29,184.96	62.60%
Cleaners	18,407.03	$8,541.62	72367.53%	8,541.62	46.40%
Bottles and Cages	64,274.95	$38,214.28	515653.01%	38,214.28	59.45%
Bike Stands	39,591.00	$24,782.97	15586.77%	24,782.97	62.60%
Bike Racks	237,096.21	$95,006.13	37778.38%	95,006.13	40.07%
Total	**109,809,274.00**	**$12,551,285.93**	**3673444.26%**	**12,551,285.93**	**11.43%**

When to use Calculated Column (CC):

```
Discount Category (CC) =
IF(
    Sales[Unit Price Discount Pct] =0, "No Discount",
    IF(Sales[Profit (CC)] > 0, "Discount with Profit", "Discount with Loss")
)
```

unt	Product Standard Cost	Total Product Cost	Sales Amount	Unit Price Discount Pct	Prof



Category	Sum of Sales Amount	Profit (CM)	Profit Margin % (CM)
Components	98,535.06	11,766.90	11.94%
Clothing	341,490.37	91,671.95	26.84%
Vests	95,633.11	28,850.62	30.17%
Tights	36,840.06	9,154.65	24.85%
Socks	6,961.37	2,177.83	31.28%
Shorts	122,228.30	33,369.20	27.30%
Gloves	65,836.88	14,597.09	22.17%
Bib-Shorts	13,990.65	3,522.56	25.18%
Bikes	860,233.82	29,910.79	3.48%
Mountain Bikes	860,233.82	29,910.79	3.48%
Accessories	143,811.43	40,313.40	28.03%
Tires and Tubes	15.62	5.34	34.19%
Pumps	693.11	190.11	27.43%
Locks	1,165.22	319.58	27.43%
Hydration Packs	17,677.52	5,790.12	32.75%
Helmets	58,584.78	13,716.80	23.41%
Cleaners	3,365.63	1,073.16	31.89%
Bottles and Cages	2,165.34	677.84	31.30%
Bike Racks	60,144.21	18,540.45	30.83%
Total	1,444,070.68	173,663.04	12.03%

Discount Category (CC)
- ○ Discount with Loss
- ● Discount with Profit
- ○ No Discount

Calculated Column

- Slice or filter values
- Categorize texts/numbers
- Expressions that are strictly bound to current row
- Consume Memory
- Calculated at refresh time
- Data can be seen in the Data tab

Calculated Measures

- Calculate percentages
- Calculate Ratios
- Complex aggregations
- Consume CPU
- Calculated on the fly

DIVIDE()
Total Cost (CM) = SUM(Sales[Total Product Cost])
Mark UP (CM) = DIVIDE([Profit (CM)],[Total Cost (CM)],0)

Category	Sum of Sales Amount	Profit (CM)	Profit Margin % (CM)	Mark Up % (CM)
⊟ Accessories	**1,272,059.71**	**634,391.75**	**49.87%**	**99.49%**
Bike Racks	237,096.21	95,006.13	40.07%	66.86%
Bike Stands	39,591.00	24,782.97	62.60%	167.36%
Bottles and Cages	64,274.95	38,214.28	59.45%	146.64%
Cleaners	18,407.03	8,541.62	46.40%	86.58%
Fenders	46,619.58	29,184.96	62.60%	167.40%
Helmets	484,049.97	226,383.23	46.77%	87.86%
Hydration Packs	105,826.62	49,040.07	46.34%	86.36%
Locks	16,225.22	5,025.51	30.97%	44.87%
Pumps	13,514.71	4,196.75	31.05%	45.04%
Tires and Tubes	246,454.42	154,016.23	62.49%	166.62%
⊟ Bikes	**94,620,526.59**	**10,515,090.51**	**11.11%**	**12.50%**
Mountain Bikes	36,445,444.52	5,932,911.85	16.28%	19.44%
Road Bikes	43,878,791.78	4,364,897.96	9.95%	11.05%
Touring Bikes	14,296,290.29	217,280.70	1.52%	1.54%
⊟ Clothing	**2,117,613.52**	**368,846.60**	**17.42%**	**21.09%**
Bib-Shorts	166,739.64	51,257.58	30.74%	44.39%
Caps	51,230.10	-1,198.49	-2.34%	-2.29%
Gloves	242,796.58	83,411.82	34.35%	52.33%
Jerseys	752,257.82	-92,911.64	-12.35%	-10.99%
Shorts	413,522.68	155,941.17	37.71%	60.54%
Socks	29,745.21	12,164.99	40.90%	69.20%
Tights	201,832.98	60,715.09	30.08%	43.02%
Vests	259,488.51	99,466.08	38.33%	62.16%
⊞ Components	**11,799,074.18**	**1,032,957.07**	**8.75%**	**9.59%**
Total	**109,809,274.00**	**12,551,285.93**	**11.43%**	**12.91%**

COUNT(), COUNTROWS, DISTINCTCOUNT:

Total No of Product (CM) = COUNT('Product'[ProductKey])

Total No. of Sales (CM) = COUNTROWS(Sales)

Total No of Product Sold (CM) = DISTINCTCOUNT(Sales[ProductKey])

Category	Total No of Product (CM)
Accessories	35
Bikes	125
Clothing	48
Components	189
Total	**397**

Category	Total No of Product (CM)	Total No. of Sales (CM)
Accessories	35	41193
Bikes	125	40005
Clothing	48	21368
Components	189	18687
Total	**397**	**121253**

Category	Total No of Product (CM)	Total No of Product Sold (CM)	Total No. of Sales (CM)
Accessories	35	30	41193
Bikes	125	125	40005
Clothing	48	45	21368
Components	189	150	18687
Total	**397**	**350**	**121253**

Min, Max and Average

Min Unit Price (CM) = MIN(Sales[Unit Price])

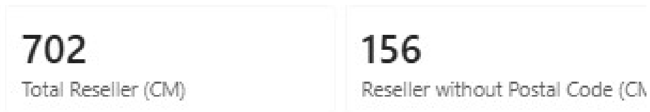
Average Unit Price (CM) = AVERAGE(Sales[Unit Price])

Max Unit Price (CM) = MAX(Sales[Unit Price])

Category	Min Unit Price (CM)	Average Unit Price (CM)	Max Unit Price (CM)
Bikes	$113.00	$1,255.08	$3,578.27
Components	$11.74	$251.40	$858.9
Clothing	$4.32	$32.09	$69.99
Accessories	$1.33	$19.69	$159
Total	**$1.33**	**$465.18**	**$3,578.27**

COUNTBLANKS

Total Reseller (CM) = COUNT(Reseller[ResellerKey])
Reseller without Postal Code (CM) = COUNTBLANK(Reseller[Postal Code])



	Filter Context	Row Context
Context creation	Provided by coordinates of the visual	Exists in Calculated Column or iterative DAX functions
Role of CALCULATE()	CALCULATE() function can be used to alter a filter context.	CALCULATE() can convert a row context to a filter context.
Relationships	Automatically follow the relationships from the 1 to many side.	Does not follow relationships or create a filter context.

📄 **Row Context**

- **What it is:**
  Row context means that when DAX evaluates an expression, it looks at one row at a time.
- **Where it happens:**
  - In **calculated columns** → each row is evaluated separately.
  - In **iterators** (like FILTER(), SUMX(), AVERAGEX(), ADDCOLUMNS()) → the function goes row by row through a table, creating a row context for each row.
- **Important to know:**
  - Row context does **not** automatically follow relationships between tables.
  - Row context does **not** create a filter context by itself.
  - To turn row context into filter context, you need special functions (like CALCULATE).
- **Measures vs. Row Context:**
  - **Calculated columns** and **iterators** have row context.
  - **Regular measures** do not have row context because they work on entire columns/tables, not individual rows.
  - If you want row-by-row evaluation inside a measure, you must use an iterator (like SUMX).

✅ **In short:**
Row context = "DAX is looking at one row at a time."
It exists in calculated columns and iterators, but not in regular measures unless you use an iterator.

📄 **Filter Context**

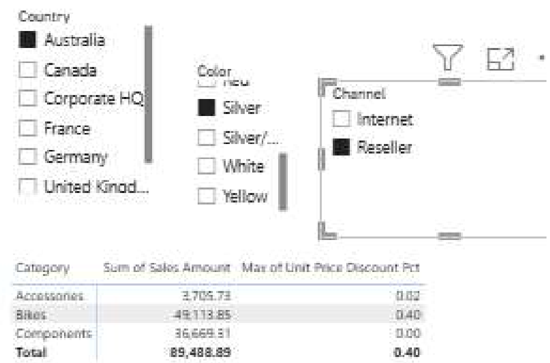
- **What it is:**
  Filter context means the set of filters applied to your data model when DAX evaluates an expression. It defines *which rows of data are visible* at the time of calculation.
- **Where it comes from (initial filter context):**
  ✓ **Slicers** on a report

✓ **Fields in visuals** (e.g., rows and columns in a Matrix)
✓ **Page or report filters**
✓ **Other visuals** that cross-filter

- **How it behaves:**
  - Filters automatically follow relationships in the data model (from the "one" side to the "many" side).
  - You can **modify or replace filter context** using the CALCULATE() function, which changes the filters before evaluating the expression.

✅ **In short:**
Filter context = "Which rows are included in the calculation."
It comes from slicers, visuals, and filters, and can be changed with CALCULATE().

Initial or incoming filter context



The Unit price discount was applicable for purchases through resellers only

📄 **Evaluation Context in Functions**
Different DAX functions change or use **evaluation context** in specific ways. Here are some common examples:

- **RELATED()** → Expands the current row context to bring in values from another table (like a VLOOKUP).
- **FILTER()** → Creates a new filter context by selecting only a subset of rows that meet certain conditions.
- **ALL()** → Removes all filters, so calculations are done on the entire table or column.
- **ALLEXCEPT()** → Removes all filters *except* the ones you specify, letting you keep certain filters while ignoring others.
- **EARLIER() / EARLIEST()** → Used in calculated columns to reference values from an outer row context when working inside nested row evaluations (like looping through rows).

RELATED()
Current List Price (CC) = RELATED('Product'[List Price])

Profit (CC)	Profit Margin % (CC)	Discount Category (CC)	Current List Price (CC)
$6.31	1.50%	No Discount	699.0982
$6.31	1.50%	No Discount	699.0982
$6.31	1.50%	No Discount	699.0982
$6.31	1.50%	No Discount	699.0982
$6.31	1.50%	No Discount	699.0982
$6.31	1.50%	No Discount	699.0982
$6.31	1.50%	No Discount	699.0982
$6.31	1.50%	No Discount	699.0982
$6.31	1.50%	No Discount	699.0982
$6.31	1.50%	No Discount	699.0982
$6.31	1.50%	No Discount	699.0982
$6.31	1.50%	No Discount	699.0982
$6.31	1.50%	No Discount	699.0982

Relatedtable()
No of Sales (CC) = COUNTROWS(RELATEDTABLE(Sales))

gin % (CC)	Discount Category (CC)	Current List Price (CC)	No of Sales (CC)
1.50%	No Discount	699.0982	121253
1.50%	No Discount	699.0982	121253
1.50%	No Discount	699.0982	121253
1.50%	No Discount	699.0982	121253
1.50%	No Discount	699.0982	121253
1.50%	No Discount	699.0982	121253
1.50%	No Discount	699.0982	121253
1.50%	No Discount	699.0982	121253
1.50%	No Discount	699.0982	121253
1.50%	No Discount	699.0982	121253
1.50%	No Discount	699.0982	121253

count Category (CC)	Current List Price (CC)	No of Sales (CC)
Discount	699.0982	1
Discount	699.0982	1
Discount	699.0982	1
Discount	699.0982	1
Discount	699.0982	1

DISTINCTCOUNT()

No of Sales (CC) = COUNTROWS(RELATEDTABLE(Sales))
Last Order Date (CC) = CALCULATE(MAX(Sales[OrderDate]))
No of Product purchased (CC) = DISTINCTCOUNT(Sales[ProductKey])
No of Product purchased CAL (CC) = CALCULATE(DISTINCTCOUNT(Sales[ProductKey]))

ı	No of Product purchased (CC)	No of Product purchased CAL (CC)	Last Order Date (CC)	No of Sales (CC)
ary 8, 1947	350	1	6/11/2020	1
er 11, 1954	350	1	10/17/2019	1
une 3, 1959	350	1	1/19/2020	1
er 29, 2002	350	1	8/18/2019	1
rch 1, 1944	350	1	11/11/2019	1
ly 17, 1954	350	2	10/20/2019	2
er 30, 1951	350	2	10/1/2019	2
er 16, 1969	350	2	4/27/2020	2
er 12, 1948	350	2	7/2/2019	2
ary 2, 1958	350	2	12/23/2019	2
pril 6, 1994	350	2	10/3/2019	2
ry 19, 1964	350	2	3/11/2020	2

VALUE()

Testing _CT _ VALUES_ Product (CT) = VALUES('Product') // Whole Table

roductKey	Product	Standard Cost	Color	List Price	Model	Subcategory	Category	SKU
376	Road-250 Black, 48	1554.9479	Black	2443.35	Road-250	Road Bikes	Bikes	BK-R89B-48
378	Road-250 Black, 52	1554.9479	Black	2443.35	Road-250	Road Bikes	Bikes	BK-R89B-52
380	Road-250 Black, 58	1554.9479	Black	2443.35	Road-250	Road Bikes	Bikes	BK-R89B-58
374	Road-250 Black, 44	1554.9479	Black	2443.35	Road-250	Road Bikes	Bikes	BK-R89B-44
606	Road-750 Black, 52	343.6496	Black	539.99	Road-750	Road Bikes	Bikes	BK-R19B-52

Testing _CT _ VALUES_ Product (CT) = VALUES('Product'[Category]) // Specific Column

```
1 Testing _CT _ VALUES_ Product (CT) = VALUES('Product'[Category])
```

Category
Clothing
Bikes
Components
Accessories

DISTINCT()

Testing _CT _ VALUES_ Product (CT) = DISTINCT('Product'[Category])

Structure | Relationships | Calculations | Calendars

```
1 Testing _CT _ VALUES_ Product (CT) = DISTINCT('Product'[Category])
```

Category
Clothing
Bikes
Components
Accessories

ALL()

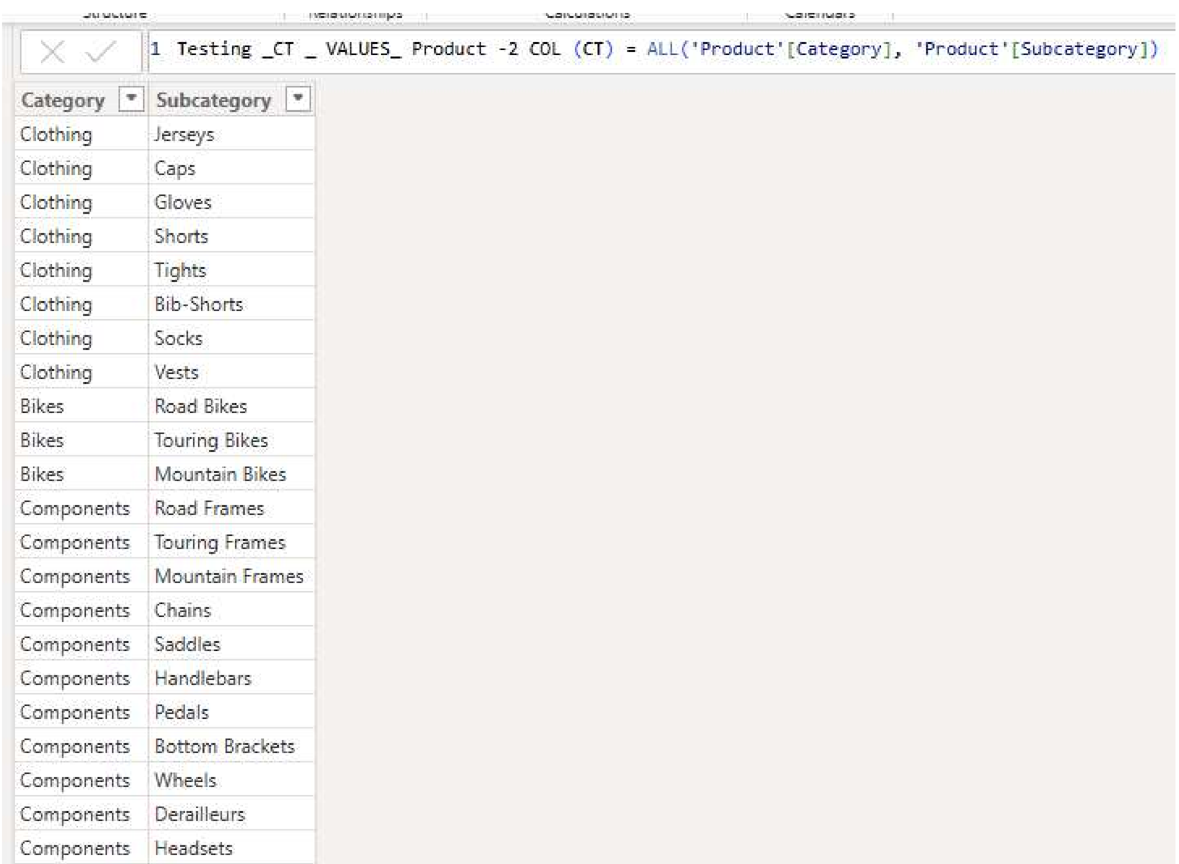
Testing _CT _ VALUES_ Product (CT) = ALL('Product'[Category])

```
1 Testing _CT _ VALUES_ Product (CT) = ALL('Product'[Category])
```

Category
Clothing
Bikes
Components
Accessories

ALL() – 2 Column

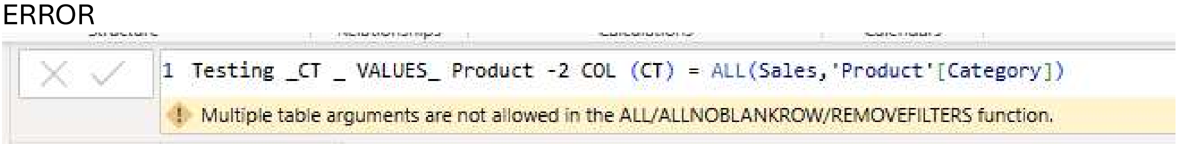
Testing _CT _ VALUES_ Product -2 COL (CT) = ALL('Product'[Category], 'Product'[Subcategory])



```
1 Testing _CT _ VALUES_ Product -2 COL (CT) = ALL('Product'[Category], 'Product'[Subcategory])
```

Category	Subcategory
Clothing	Jerseys
Clothing	Caps
Clothing	Gloves
Clothing	Shorts
Clothing	Tights
Clothing	Bib-Shorts
Clothing	Socks
Clothing	Vests
Bikes	Road Bikes
Bikes	Touring Bikes
Bikes	Mountain Bikes
Components	Road Frames
Components	Touring Frames
Components	Mountain Frames
Components	Chains
Components	Saddles
Components	Handlebars
Components	Pedals
Components	Bottom Brackets
Components	Wheels
Components	Derailleurs
Components	Headsets

ERROR

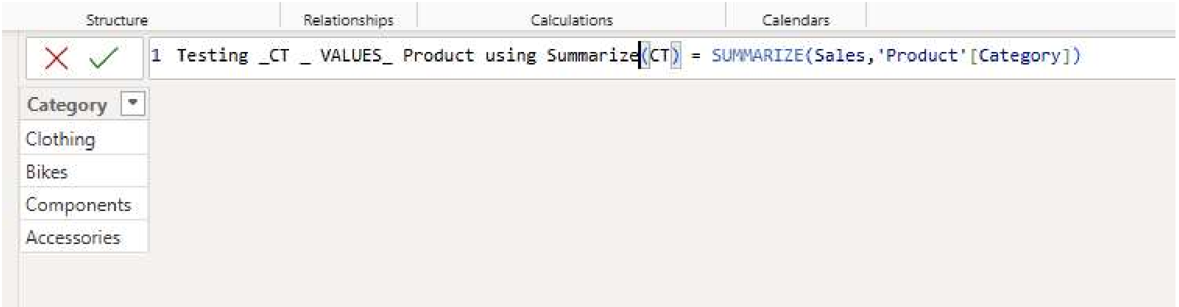


```
1 Testing _CT _ VALUES_ Product -2 COL (CT) = ALL(Sales,'Product'[Category])
```

Multiple table arguments are not allowed in the ALL/ALLNOBLANKROW/REMOVEFILTERS function.

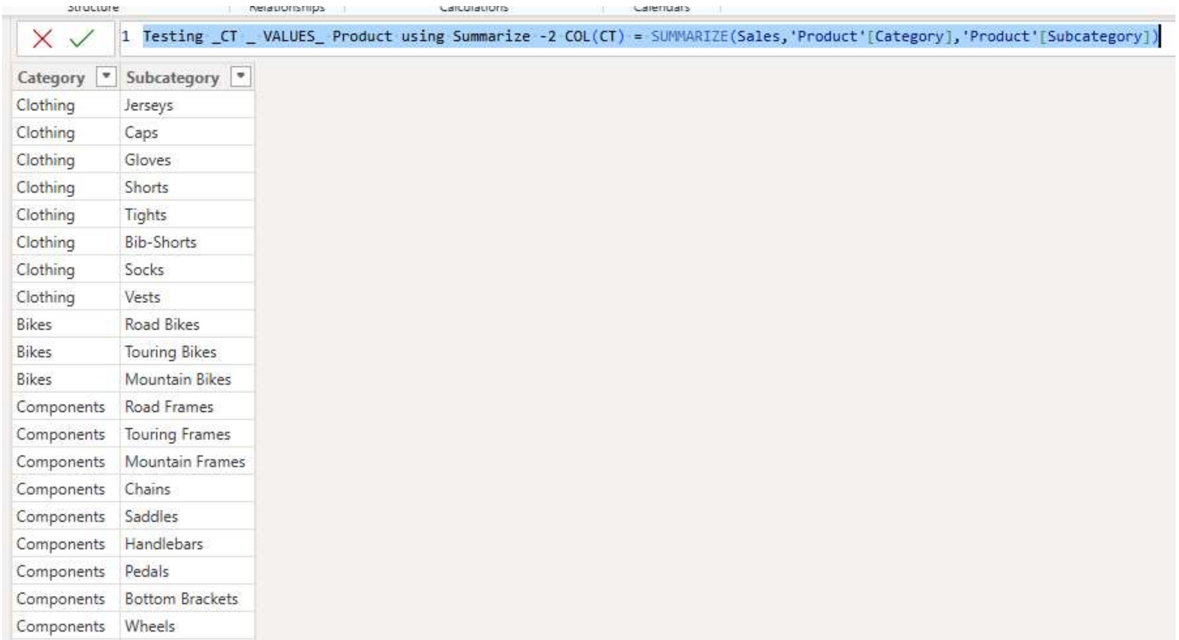
SUMMARIZE()

Testing _CT _ VALUES_ Product using Summarize(CT) = SUMMARIZE(Sales,'Product'[Category])



Category
Clothing
Bikes
Components
Accessories

Testing _CT _ VALUES_ Product using Summarize -2 COL(CT) =
SUMMARIZE(Sales,'Product'[Category],'Product'[Subcategory])



Category	Subcategory
Clothing	Jerseys
Clothing	Caps
Clothing	Gloves
Clothing	Shorts
Clothing	Tights
Clothing	Bib-Shorts
Clothing	Socks
Clothing	Vests
Bikes	Road Bikes
Bikes	Touring Bikes
Bikes	Mountain Bikes
Components	Road Frames
Components	Touring Frames
Components	Mountain Frames
Components	Chains
Components	Saddles
Components	Handlebars
Components	Pedals
Components	Bottom Brackets
Components	Wheels

SUMMARIZE() using measure or expression for GroupBy

Testing _CT _ VALUES_ Product using Summarize -using Measures(CT) =
SUMMARIZE(
 Sales,
'Product'[Category],
'Product'[Subcategory],
"Sales",SUM(Sales[Sales Amount]),
"Total Cost", SUM(Sales[Total Product Cost])
)

```
Testing _CT _ VALUES_ Product using Summarize -using Measures(CT) =
SUMMARIZE(
 Sales,
'Product'[Category],
'Product'[Subcategory],
"Sales",SUM(Sales[Sales Amount]),
"Total Cost", SUM(Sales[Total Product Cost])
)
```

Category	Subcategory	Sales	Total Cost
Clothing	Jerseys	752257.819999984	845169.459999948
Clothing	Caps	51230.1000000001	52428.59
Clothing	Gloves	242796.579999997	159384.76
Clothing	Shorts	413522.679999999	257581.509999999
Clothing	Tights	201832.980000002	141117.89
Clothing	Bib-Shorts	166739.640000002	115482.06
Clothing	Socks	29745.21	17580.22
Clothing	Vests	259488.509999999	160022.430000001
Bikes	Road Bikes	43878791.7800045	39513893.8199937
Bikes	Touring Bikes	14296290.2899999	14079009.5899992
Bikes	Mountain Bikes	36445444.5199996	30512532.6700002
Components	Road Frames	3849853.72000003	3710985.94999996
Components	Touring Frames	1642327.49000001	1647899.57
Components	Mountain Frames	4713671.55999999	4225828.71999997
Components	Chains	9377.60999999999	6955.73
Components	Saddles	55828.7499999998	41352.0199999999
Components	Handlebars	170589.93	126373.06

SUMMARIZE() to get data for all categories:

Testing _CT _ VALUES_ Product using Summarize -all category(CT) =
SUMMARIZE(
 'Product',
'Product'[Category],
'Product'[Subcategory]
)

Structure | Relationships | Calculations | Calendars

```
Testing _CT _ VALUES_ Product using Summarize -all category(CT) =
SUMMARIZE(
 'Product',
'Product'[Category],
'Product'[Subcategory]
)
```

Category	Subcategory
Clothing	Jerseys
Clothing	Caps
Clothing	Gloves
Clothing	Shorts
Clothing	Tights
Clothing	Bib-Shorts
Clothing	Socks
Clothing	Vests
Bikes	Road Bikes
Bikes	Touring Bikes
Bikes	Mountain Bikes
Components	Road Frames
Components	Touring Frames
Components	Mountain Frames
Components	Chains
Components	Saddles
Components	Handlebars
Components	Pedals
Components	Bottom Brackets

Total Sales Amount (CM) = SUM(Sales[Sales Amount])

Testing _CT _ VALUES_ Product using Summarize -all category(CT) =
SUMMARIZE(
 'Product',
'Product'[Category],
'Product'[Subcategory],
"SALES", [Total Sales Amount (CM)]
)

```
Testing _CT _ VALUES_ Product using Summarize -all category(CT) =
SUMMARIZE(
'Product',
'Product'[Category],
'Product'[Subcategory],
"SALES", [Total Sales Amount (CM)]
)
```

Category	Subcategory	SALES
Accessories	Lights	
Accessories	Panniers	
Components	Chains	9378
Accessories	Pumps	13515
Accessories	Locks	16225
Accessories	Cleaners	18407
Clothing	Socks	29745
Accessories	Bike Stands	39591
Accessories	Fenders	46620
Clothing	Caps	51230
Components	Bottom Brackets	51826
Components	Saddles	55829
Components	Headsets	60942
Accessories	Bottles and Cages	64275
Components	Brakes	66019
Components	Derailleurs	70209
Components	Forks	77932
Accessories	Hydration Packs	105827

ALLEXCEPT

Testing _CT _ VALUES_ Product using ALLEXCEPT = ALLEXCEPT('Product', 'Product'[ProductKey])

```
Testing _CT _ VALUES_ Product using ALLEXCEPT = ALLEXCEPT('Product', 'Product'[ProductKey])
```

Product	Standard Cost	Color	List Price	Model	Subcategory	Category	SKU
Road-250 Black, 48	1554.9479	Black	2443.35	Road-250	Road Bikes	Bikes	BK-R89B-48
Road-250 Black, 52	1554.9479	Black	2443.35	Road-250	Road Bikes	Bikes	BK-R89B-52
Road-250 Black, 58	1554.9479	Black	2443.35	Road-250	Road Bikes	Bikes	BK-R89B-58
Road-250 Black, 44	1554.9479	Black	2443.35	Road-250	Road Bikes	Bikes	BK-R89B-44
Road-750 Black, 52	343.6496	Black	539.99	Road-750	Road Bikes	Bikes	BK-R19B-52
Road-750 Black, 44	343.6496	Black	539.99	Road-750	Road Bikes	Bikes	BK-R19B-44
Road-750 Black, 48	343.6496	Black	539.99	Road-750	Road Bikes	Bikes	BK-R19B-48
Road-750 Black, 58	343.6496	Black	539.99	Road-750	Road Bikes	Bikes	BK-R19B-58
Road-650 Black, 62	486.7066	Black	782.99	Road-650	Road Bikes	Bikes	BK-R50B-62
Road-650 Black, 58	486.7066	Black	782.99	Road-650	Road Bikes	Bikes	BK-R50B-58
Road-650 Black, 60	486.7066	Black	782.99	Road-650	Road Bikes	Bikes	BK-R50B-60
Road-650 Black, 44	486.7066	Black	782.99	Road-650	Road Bikes	Bikes	BK-R50B-44
Road-650 Black, 48	486.7066	Black	782.99	Road-650	Road Bikes	Bikes	BK-R50B-48
Road-650 Black, 52	486.7066	Black	782.99	Road-650	Road Bikes	Bikes	BK-R50B-52
LL Road Frame - Black, 58	204.6251	Black	337.22	LL Road Frame	Road Frames	Components	FR-R38B-58
LL Road Frame - Black, 60	204.6251	Black	337.22	LL Road Frame	Road Frames	Components	FR-R38B-60
LL Road Frame - Black, 62	204.6251	Black	337.22	LL Road Frame	Road Frames	Components	FR-R38B-62
LL Road Frame - Black, 44	204.6251	Black	337.22	LL Road Frame	Road Frames	Components	FR-R38B-44
LL Road Frame - Black, 48	204.6251	Black	337.22	LL Road Frame	Road Frames	Components	FR-R38B-48
LL Road Frame - Black, 52	204.6251	Black	337.22	LL Road Frame	Road Frames	Components	FR-R38B-52
HL Road Frame - Black, 44	868.6342	Black	1431.5	HL Road Frame	Road Frames	Components	FR-R92B-44

FILTER()

Testing_CT_Product using Fiter (CT) = FILTER('Product', 'Product'[Category] = "Accessories")

Structure | Relationships | Calculations | Calendars

```
1 Testing_CT_Product using Fiter (CT) = FILTER('Product', 'Product'[Category] = "Accessories")
```

ProductKey	Product	Standard Cost	Color	List Price	Model	Subcategory	Category	SKU
216	Sport-100 Helmet, Black	13.8782	Black	33.6442	Sport-100	Helmets	Accessories	HL-U509
217	Sport-100 Helmet, Black	13.0863	Black	34.99	Sport-100	Helmets	Accessories	HL-U509
215	Sport-100 Helmet, Black	12.0278	Black	33.6442	Sport-100	Helmets	Accessories	HL-U509
450	Taillights - Battery-Powered	5.7709	NA	13.99	Taillight	Lights	Accessories	LT-T990
448	Minipump	8.2459	NA	19.99	Minipump	Pumps	Accessories	PU-0452
449	Mountain Pump	10.3084	NA	24.99	Mountain Pump	Pumps	Accessories	PU-M044
451	Headlights - Dual-Beam	14.4334	NA	34.99	Headlights - Dual-Beam	Lights	Accessories	LT-H902
452	Headlights - Weatherproof	18.5584	NA	44.99	Headlights - Weatherproof	Lights	Accessories	LT-H903
446	Touring-Panniers, Large	51.5625	Grey	125	Touring-Panniers	Panniers	Accessories	PA-T100
447	Cable Lock	10.3125	NA	25	Cable Lock	Locks	Accessories	LO-C100
213	Sport-100 Helmet, Red	13.8782	Red	33.6442	Sport-100	Helmets	Accessories	HL-U509-R
221	Sport-100 Helmet, Blue	13.8782	Blue	33.6442	Sport-100	Helmets	Accessories	HL-U509-B
480	Patch Kit/8 Patches	0.8565	NA	2.29	Patch kit	Tires and Tubes	Accessories	PK-7098
529	Road Tire Tube	1.4923	NA	3.99	Road Tire Tube	Tires and Tubes	Accessories	TT-R982
528	Mountain Tire Tube	1.8663	NA	4.99	Mountain Tire Tube	Tires and Tubes	Accessories	TT-M928
477	Water Bottle - 30 oz.	1.8663	NA	4.99	Water Bottle	Bottles and Cages	Accessories	WB-H098
530	Touring Tire Tube	1.8663	NA	4.99	Touring Tire Tube	Tires and Tubes	Accessories	TT-T092
479	Road Bottle Cage	3.3623	NA	8.99	Road Bottle Cage	Bottles and Cages	Accessories	BC-R205
478	Mountain Bottle Cage	3.7363	NA	9.99	Mountain Bottle Cage	Bottles and Cages	Accessories	BC-M005
538	LL Road Tire	8.0373	NA	21.49	LL Road Tire	Tires and Tubes	Accessories	TI-R092
535	LL Mountain Tire	9.3463	NA	24.99	LL Mountain Tire	Tires and Tubes	Accessories	TI-M267
539	ML Road Tire	9.3463	NA	24.99	ML Road Tire	Tires and Tubes	Accessories	TI-R628
541	Touring Tire	10.8423	NA	28.99	Touring Tire	Tires and Tubes	Accessories	TI-T723

Multiple Filter:

Testing_CT_Product using Fiter (CT) =
FILTER('Product',
'Product'[Category] = "Accessories"
&& 'Product'[List Price]>30
)

tructure | Relationships | Calculations | Calendars

```
1 Testing_CT_Product using Fiter (CT) =
2 FILTER('Product',
3 'Product'[Category] = "Accessories"
4 && 'Product'[List Price]>30
5 )
```

tKey	Product	Standard Cost	Color	List Price	Model	Subcategory	Category	SKU
216	Sport-100 Helmet, Black	13.8782	Black	33.6442	Sport-100	Helmets	Accessories	HL-U509
217	Sport-100 Helmet, Black	13.0863	Black	34.99	Sport-100	Helmets	Accessories	HL-U509
215	Sport-100 Helmet, Black	12.0278	Black	33.6442	Sport-100	Helmets	Accessories	HL-U509
451	Headlights - Dual-Beam	14.4334	NA	34.99	Headlights - Dual-Beam	Lights	Accessories	LT-H902
452	Headlights - Weatherproof	18.5584	NA	44.99	Headlights - Weatherproof	Lights	Accessories	LT-H903
446	Touring-Panniers, Large	51.5625	Grey	125	Touring-Panniers	Panniers	Accessories	PA-T100
213	Sport-100 Helmet, Red	13.8782	Red	33.6442	Sport-100	Helmets	Accessories	HL-U509-R
221	Sport-100 Helmet, Blue	13.8782	Blue	33.6442	Sport-100	Helmets	Accessories	HL-U509-B
214	Sport-100 Helmet, Red	13.0863	Red	34.99	Sport-100	Helmets	Accessories	HL-U509-R
222	Sport-100 Helmet, Blue	13.0863	Blue	34.99	Sport-100	Helmets	Accessories	HL-U509-B
487	Hydration Pack - 70 oz.	20.5663	Silver	54.99	Hydration Pack	Hydration Packs	Accessories	HY-1023-70
483	Hitch Rack - 4-Bike	44.88	NA	120	Hitch Rack - 4-Bike	Bike Racks	Accessories	RA-H123
486	All-Purpose Bike Stand	59.466	NA	159	All-Purpose Bike Stand	Bike Stands	Accessories	ST-1401
537	HL Mountain Tire	13.09	NA	35	HL Mountain Tire	Tires and Tubes	Accessories	TI-M823
540	HL Road Tire	12.1924	NA	32.6	HL Road Tire	Tires and Tubes	Accessories	TI-R982
212	Sport-100 Helmet, Red	12.0278	Red	33.6442	Sport-100	Helmets	Accessories	HL-U509-R
220	Sport-100 Helmet, Blue	12.0278	Blue	33.6442	Sport-100	Helmets	Accessories	HL-U509-B

Virtual table lineage

CM

Imaginary Table of customer with sales greater than $5000 =
CALCULATE(COUNTROWS(Customer),
FILTER(Customer,[Total Sales Amount (CM)]>5000)
)

Customer	Imaginary Table of customer with sales greater than $5000.00	First Product	Total Cost (CM)	Total Sales Amount (CM)
Abby Raman	1	All-Purpose Bike Stand	2,887.99	5,193.72
Abby Sai	1	All-Purpose Bike Stand	4,718.96	8,262.52
Abby Sanchez	1	All-Purpose Bike Stand	3,726.24	6,021.62
Abby Sandberg	1	All-Purpose Bike Stand	3,455.61	5,948.23
Abby Subram	1	All-Purpose Bike Stand	4,276.19	7,494.88
Abigail Brooks	1	All-Purpose Bike Stand	3,436.91	5,898.26
Abigail Clark	1	All-Purpose Bike Stand	3,442.52	5,913.24
Abigail Foster	1	All-Purpose Bike Stand	3,485.53	6,039.73
Abigail Henderson	1	All-Purpose Bike Stand	3,455.06	5,958.22
Abigail Howard	1	All-Purpose Bike Stand	3,451.32	5,948.24
Abigail Richardson	1	All-Purpose Bike Stand	3,194.61	5,710.23
Adam Baker	1	All-Purpose Bike Stand	3,428.88	5,888.24
Adam Flores	1	All-Purpose Bike Stand	3,671.40	6,515.30
Adam Hill	1	All-Purpose Bike Stand	3,739.33	6,056.61
Adam Ross	1	All-Purpose Bike Stand	4,740.06	8,215.99
Adrian Ramirez	1	All-Purpose Bike Stand	3,451.68	5,937.73
Adriana Chandra	1	All-Purpose Bike Stand	4,938.86	8,326.31
Adriana Gonzalez	1	All-Purpose Bike Stand	8,005.47	13,242.70
Adriana Lopez	1	All-Purpose Bike Stand	4,670.44	8,145.33
Adriana Smith	1	All-Purpose Bike Stand	3,295.37	5,333.25
Adrienne Gomez	1	All-Purpose Bike Stand	4,674.56	8,156.32
Adrienne Sanz	1	All-Purpose Bike Stand	4,114.91	7,000.87
Adrienne Serrano	1	All-Purpose Bike Stand	4,110.69	6,999.09
Adrienne Torres	1	All-Purpose Bike Stand	4,669.58	8,143.03
Aidan Perry	1	All-Purpose Bike Stand	3,489.23	5,985.21
Aidan Ross	1	All-Purpose Bike Stand	3,464.96	5,973.24
Aimee He	1	All-Purpose Bike Stand	4,683.83	8,180.00
Aimee She	1	All-Purpose Bike Stand	3,437.77	5,900.55
Alan Chen	1	All-Purpose Bike Stand	4,288.37	7,295.04
Alan Guo	1	All-Purpose Bike Stand	3,465.14	5,964.22
Alan He	1	All-Purpose Bike Stand	3,884.12	6,825.45
Alan Huang	1	All-Purpose Bike Stand	3,871.69	6,802.13
Total	**1733**	**All-Purpose Bike Stand**	**97,257,988.07**	**109,809,274.00**

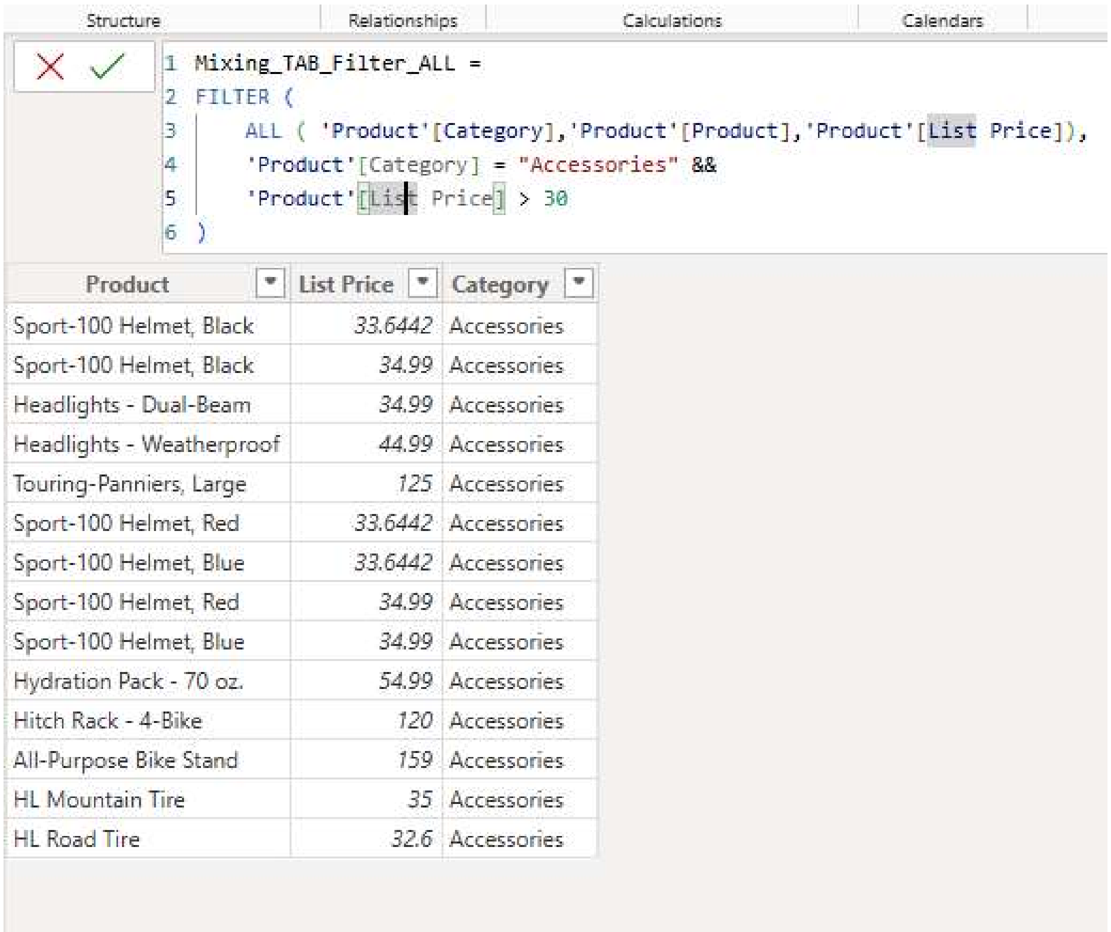
CT
Imaginary_Customer_Sales_Above5000 =
FILTER (
  Customer,
  [Total Sales Amount (CM)] >= 5000
)

```
Imaginary_Customer_Sales_Above5000 =
FILTER (
    Customer,
    [Total Sales Amount (CM)] >= 5000
)
```

erKey	Customer ID	Customer	City	State-Province	Country-Region	Postal Code	Date of Birth	No of Product purchased (CC)
14224	AW00014224	Tamara She	Lane Cove	New South Wales	Australia	1597	10/21/1981 12:00:00 AM	
18691	AW00018691	Claudia Zheng	Lane Cove	New South Wales	Australia	1597	10/26/1966 12:00:00 AM	
19592	AW00019592	Tara Xu	Darlinghurst	New South Wales	Australia	2010	6/2/1965 12:00:00 AM	
16496	AW00016496	Dawn Tang	Darlinghurst	New South Wales	Australia	2010	2/2/1946 12:00:00 AM	
18484	AW00018484	Dustin Sharma	Darlinghurst	New South Wales	Australia	2010	2/18/1967 12:00:00 AM	
11110	AW00011110	Curtis Yang	Darlinghurst	New South Wales	Australia	2010	6/6/1982 12:00:00 AM	
11930	AW00011930	Jaclyn Nara	North Sydney	New South Wales	Australia	2055	6/21/1960 12:00:00 AM	
19568	AW00019568	Candice Sun	North Sydney	New South Wales	Australia	2055	7/11/1987 12:00:00 AM	
13650	AW00013650	Jessie She	North Sydney	New South Wales	Australia	2055	3/5/1977 12:00:00 AM	
14263	AW00014263	Nina Lal	Silverwater	New South Wales	Australia	2264	2/1/1994 12:00:00 AM	
18210	AW00018210	Katie Lal	Silverwater	New South Wales	Australia	2264	9/9/1941 12:00:00 AM	
18207	AW00018207	Phillip Gonzalez	Silverwater	New South Wales	Australia	2264	5/3/1997 12:00:00 AM	
12349	AW00012349	Lacey Yuan	Silverwater	New South Wales	Australia	2264	11/14/1959 12:00:00 AM	
12346	AW00012346	Wesley Gao	Silverwater	New South Wales	Australia	2264	11/22/1972 12:00:00 AM	
11445	AW00011445	Kari Kim	Silverwater	New South Wales	Australia	2264	10/12/1948 12:00:00 AM	
20448	AW00020448	Jésus Blanco	Springwood	New South Wales	Australia	2777	12/15/1945 12:00:00 AM	
20623	AW00020623	Ruth Garcia	Springwood	New South Wales	Australia	2777	11/24/1980 12:00:00 AM	
13135	AW00013135	Johnny Rai	Springwood	New South Wales	Australia	2777	11/21/1942 12:00:00 AM	
13128	AW00013128	Colleen West	Springwood	New South Wales	Australia	2777	3/4/1955 12:00:00 AM	

Mixing_TAB_Filter_ALL (CT)

```
Mixing_TAB_Filter_ALL =
FILTER (
   ALL ( 'Product'[Category],'Product'[Product],'Product'[List Price]),
   'Product'[Category] = "Accessories" &&
   'Product'[List Price] > 30
)
```




```
1 Mixing_TAB_Filter_ALL =
2 FILTER (
3     ALL ( 'Product'[Category],'Product'[Product],'Product'[List Price]),
4     'Product'[Category] = "Accessories" &&
5     'Product'[List Price] > 30
6 )
```

Product	List Price	Category
Sport-100 Helmet, Black	33.6442	Accessories
Sport-100 Helmet, Black	34.99	Accessories
Headlights - Dual-Beam	34.99	Accessories
Headlights - Weatherproof	44.99	Accessories
Touring-Panniers, Large	125	Accessories
Sport-100 Helmet, Red	33.6442	Accessories
Sport-100 Helmet, Blue	33.6442	Accessories
Sport-100 Helmet, Red	34.99	Accessories
Sport-100 Helmet, Blue	34.99	Accessories
Hydration Pack - 70 oz.	54.99	Accessories
Hitch Rack - 4-Bike	120	Accessories
All-Purpose Bike Stand	159	Accessories
HL Mountain Tire	35	Accessories
HL Road Tire	32.6	Accessories

Mixing_TAB_Summarize_Filter (CT) =

```
Mixing_TAB_Summarize_Filter (CT) =
FILTER(
   SUMMARIZE(
      'Product',
      'Product'[Category],
      'Product'[Subcategory],
      "Sales", [Total Sales Amount (CM)]
   ),
   'Product'[Category]= "Clothing"
   && [Total Sales Amount (CM)]> 250000
      )
```

Structure | Formatting

```
Mixing_TAB_Summarize_Filter (CT) =
FILTER(
    SUMMARIZE(
        'Product',
        'Product'[Category],
        'Product'[Subcategory],
        "Sales", [Total Sales Amount (CM)]
    ),
    'Product'[Category]= "Clothing"
    && [Total Sales Amount (CM)]> 250000
        )
```

Category	Subcategory	Sales
Clothing	Jerseys	$752,257.82
Clothing	Shorts	$413,522.68
Clothing	Vests	$259,488.51

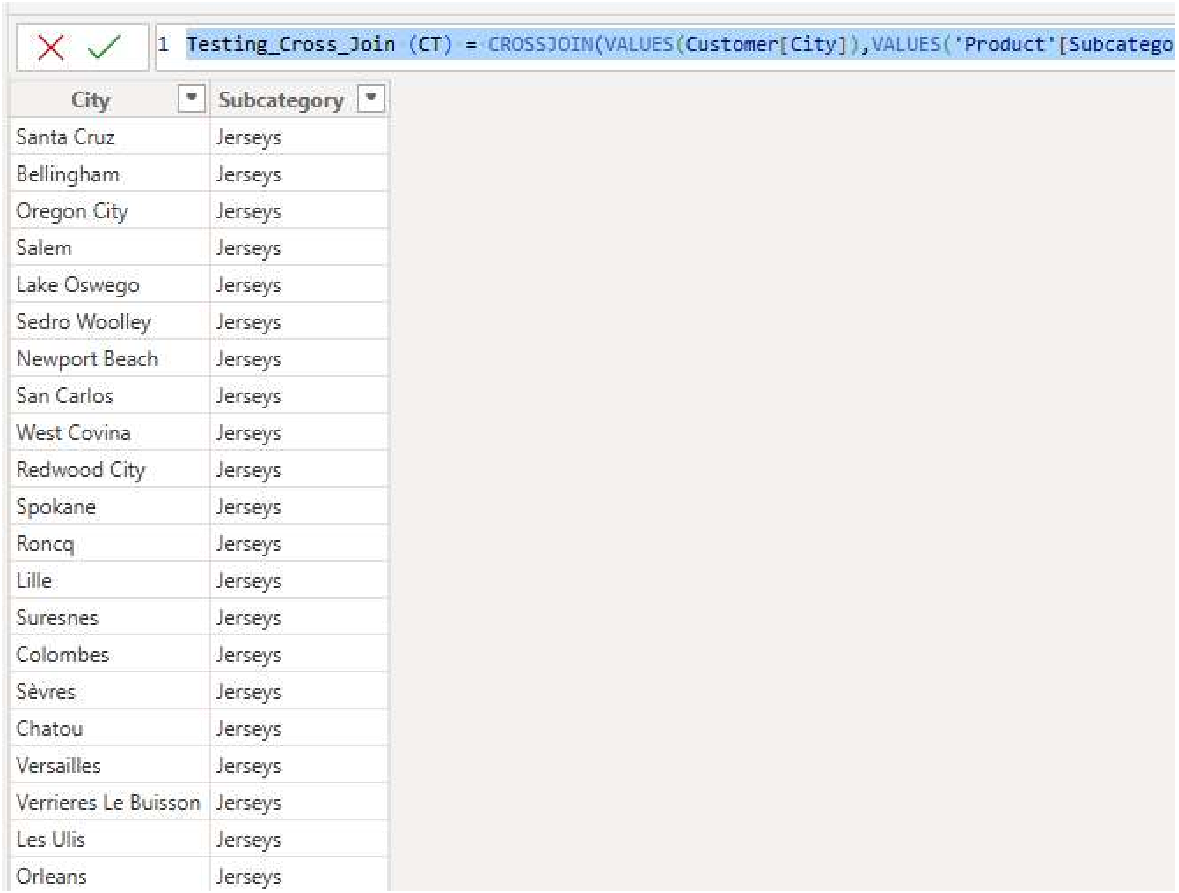
CROSS JOIN(tablee1,table2)

Testing_Cross_Join (CT) = CROSSJOIN(Customer,'Product')

```
Testing_Cross_Join (CT) = CROSSJOIN(Customer,'Product')
```

CustomerKey	Customer ID	Customer	City	State-Province	Country-Region	Postal Code	Date of Birth	No of Product purcha
14562	AW00014562	Bryce Stewart	Concord	California	United States	94519	7/17/1954 12:00:00 AM	
14919	AW00014919	Erik Torres	Concord	California	United States	94519	11/30/1951 12:00:00 AM	
15177	AW00015177	Jorge Lu	Concord	California	United States	94519	9/16/1969 12:00:00 AM	
15319	AW00015319	Jade Sanchez	Concord	California	United States	94519	10/12/1948 12:00:00 AM	
15858	AW00015858	Jonathan Parker	Concord	California	United States	94519	1/2/1958 12:00:00 AM	
17455	AW00017455	Dylan Hayes	Concord	California	United States	94519	4/6/1994 12:00:00 AM	
18328	AW00018328	Alvin Rai	Concord	California	United States	94519	2/19/1964 12:00:00 AM	
17145	AW00017145	Sarah Bennett	Concord	California	United States	94519	11/8/1982 12:00:00 AM	
17447	AW00017447	Sydney Lewis	Concord	California	United States	94519	5/19/1976 12:00:00 AM	
17562	AW00017562	Zachary Sharma	Concord	California	United States	94519	2/26/1984 12:00:00 AM	
21341	AW00021341	Louis Liu	Concord	California	United States	94519	4/12/2001 12:00:00 AM	
21449	AW00021449	Devon Raheem	Concord	California	United States	94519	11/27/1977 12:00:00 AM	

Testing_Cross_Join using Values (CT) = CROSSJOIN(VALUES(Customer[City]),VALUES('Product'[Subcategory]))



1 Testing_Cross_Join (CT) = CROSSJOIN(VALUES(Customer[City]),VALUES('Product'[Subcatego

City	Subcategory
Santa Cruz	Jerseys
Bellingham	Jerseys
Oregon City	Jerseys
Salem	Jerseys
Lake Oswego	Jerseys
Sedro Woolley	Jerseys
Newport Beach	Jerseys
San Carlos	Jerseys
West Covina	Jerseys
Redwood City	Jerseys
Spokane	Jerseys
Roncq	Jerseys
Lille	Jerseys
Suresnes	Jerseys
Colombes	Jerseys
Sèvres	Jerseys
Chatou	Jerseys
Versailles	Jerseys
Verrieres Le Buisson	Jerseys
Les Ulis	Jerseys
Orleans	Jerseys

Summarize, Vaues, Cross join

Testing Cross join, Summarize,Values (CT) =
SUMMARIZE(CROSSJOIN(VALUES(Customer[City]),VALUES('Product'[Subcategory])),
Customer[City],
'Product'[Subcategory],
"Sales", [Total Sales Amount (CM)])

Auto recovery contains some recovered files that haven't been opened.

```
1 Testing Cross jin, Summarize,Values (CT) =
2 SUMMARIZE(CROSSJOIN(VALUES(Customer[City]),VALUES('Product'[Subcategory])),
3 Customer[City],
4 'Product'[Subcategory],
5 "Sales", [Total Sales Amount (CM)])
```

City	Subcategory	Sales
Perth	Hydration Packs	329.94
Findon	Hydration Packs	439.92
Cloverdale	Hydration Packs	274.95
Paris	Hydration Packs	384.93
Saint Ouen	Hydration Packs	109.98
Cergy	Hydration Packs	109.98
Metz	Hydration Packs	54.99
Saint Germain en La	Hydration Packs	54.99
Paris La Defense	Hydration Packs	54.99
Courbevoie	Hydration Packs	109.98
Roubaix	Hydration Packs	109.98
Drancy	Hydration Packs	54.99
Bobigny	Hydration Packs	109.98
Tremblay-en-France	Hydration Packs	54.99
Saint-Denis	Hydration Packs	54.99
Münster	Hydration Packs	54.99
Offenbach	Hydration Packs	54.99
Saarlouis	Hydration Packs	109.98

Testing Table in Measure (CM)

Testing Table using Measures = COUNTROWS(Values('Product'))

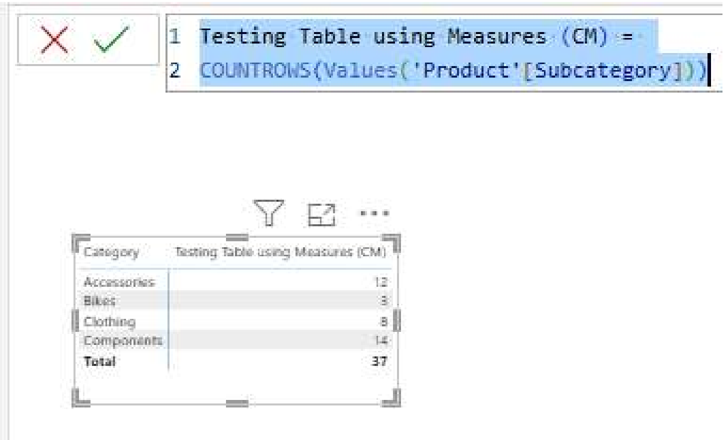
397
Testing Table using Measures

Category	Testing Table using Measures
Accessories	35
Bikes	125
Clothing	48
Components	189
Total	**397**

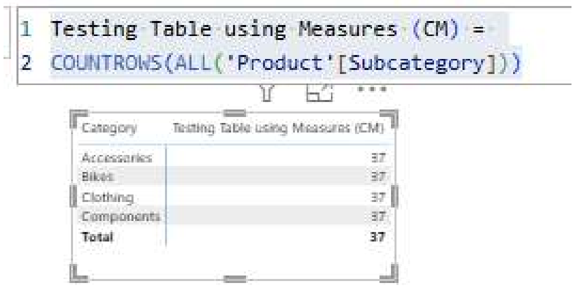
Testing Table using Measures (CM) = COUNTROWS(Values('Product'[Product]))

Category	Testing Table using Measures (CM)
Accessories	29
Bikes	97
Clothing	35
Components	134
Total	**295**

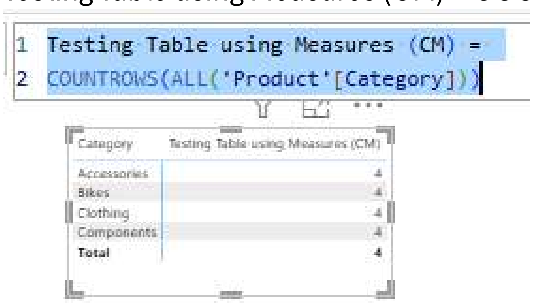
Testing Table using Measures (CM) = COUNTROWS(Values('Product'[Subcategory]))



Testing Table using Measures (CM) = COUNTROWS(ALL('Product'[Subcategory]))



Testing Table using Measures (CM) = COUNTROWS(ALL('Product'[Category]))



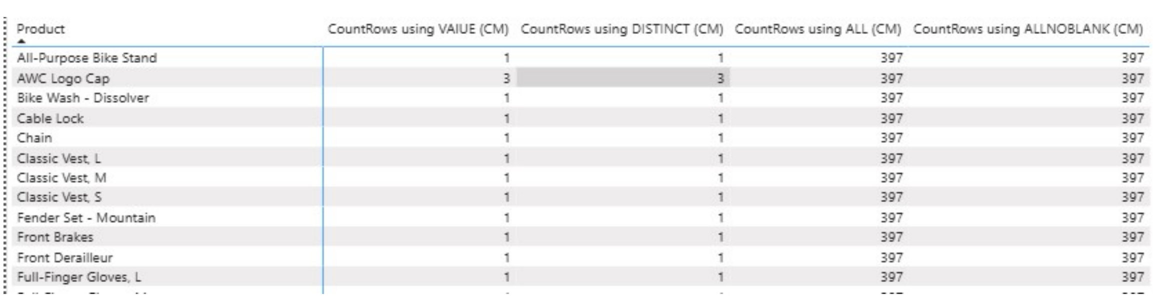
VALUES(), DISTINCT(), ALL() and ALLNOBLANKROW()

CountRows using VAlUE (CM) = COUNTROWS(VALUES('Product'))

CountRows using DISTINCT (CM) = COUNTROWS(DISTINCT('Product'))

CountRows using ALL (CM) = COUNTROWS(ALL('Product'))

CountRows using ALLNOBLANK (CM) = COUNTROWS(ALLNOBLANKROW('Product'))



Product	CountRows using VAlUE (CM)	CountRows using DISTINCT (CM)	CountRows using ALL (CM)	CountRows using ALLNOBLANK (CM)
All-Purpose Bike Stand	1	1	397	397
AWC Logo Cap	3	3	397	397
Bike Wash - Dissolver	1	1	397	397
Cable Lock	1	1	397	397
Chain	1	1	397	397
Classic Vest, L	1	1	397	397
Classic Vest, M	1	1	397	397
Classic Vest, S	1	1	397	397
Fender Set - Mountain	1	1	397	397
Front Brakes	1	1	397	397
Front Derailleur	1	1	397	397
Full-Finger Gloves, L	1	1	397	397

Variables and Comments in DAX

Testing Variables (CM) =

VAR TotalQuantity = SUM(Sales[Order Quantity])

RETURN
TotalQuantity

Category	Sum of Order Quantity	Testing Variables (CM)
Accessories	61931	61931
Bikes	90220	90220
Clothing	73598	73598
Components	49027	49027
Total	**274776**	**274776**

Testing Variables (CM) =
VAR TotalQuantity = SUM(Sales[Order Quantity])
RETURN
IF(TotalQuantity >1000, TotalQuantity*0.95, TotalQuantity*1.25)

```
Testing Variables (CM) =
VAR TotalQuantity = SUM(Sales[Order Quantity])
RETURN
IF(TotalQuantity >1000, TotalQuantity*0.95, TotalQuantity*1.25)
```

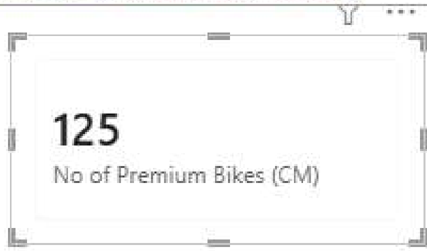
Category	Sum of Order Quantity	Testing Variables (CM)
Accessories	61931	58834
Bikes	90220	85709
Clothing	73598	69918
Components	49027	46576
Total	**274776**	**261037**

Category	Sum of Order Quantity	Testing Variables (CM)
Accessories	**61931**	**58834**
All-Purpose Bike Stand	249	311
Bike Wash	3319	3153
Cable Lock	1086	1032
Fender Set - Mountain	2121	2015
Hitch Rack - 4-Bike	3166	3008
HL Mountain Tire	1396	1326
HL Road Tire	858	1073
Hydration Pack	2761	2623
LL Mountain Tire	862	1078
LL Road Tire	1044	992
Minipump	1130	1074
ML Mountain Tire	1161	1103
ML Road Tire	926	1158
Mountain Bottle Cage	2025	1924
Mountain Tire Tube	3095	2940
Patch kit	3865	3672
Road Bottle Cage	1712	1626
Road Tire Tube	2376	2257
Total	**274776**	**261037**

No of Premium Bikes (CM) =
VAR PremiumBikeListPrice = 3000
VAR BikeCategoryProduct =
FILTER('Product','Product'[Category]="Bikes")
VAR PremiumBikeCategoryProducts =
FILTER(BikeCategoryProduct,'Product'[List Price] > PremiumBikeListPrice)
RETURN
COUNTROWS(PremiumBikeCategoryProducts)

```
No of Premium Bikes (CM) =
VAR PremiumBikeListPrice = 3000
VAR BikeCategoryProduct =
FILTER('Product','Product'[Category]="Bikes")
VAR PremiumBikeCategoryProducts =
FILTER(BikeCategoryProduct,'Product'[List Price] > PremiumBikeListPrice)
RETURN
COUNTROWS(PremiumBikeCategoryProducts)
```


```
No of Premium Bikes (CM) =
VAR PremiumBikeListPrice = 3000
VAR BikeCategoryProduct =
FILTER('Product','Product'[Category]="Bikes")
VAR PremiumBikeCategoryProducts =
FILTER(BikeCategoryProduct,'Product'[List Price] > PremiumBikeListPrice)
RETURN
COUNTROWS(BikeCategoryProduct)
```



Iterators

Discount Amount (CM) =
SUMX(Sales,Sales[Unit Price Discount Pct] * Sales[Extended Amount])

Discount Amount Wrong (CM) =
SUM(Sales[Unit Price Discount Pct]) * SUM(Sales[Extended Amount])

```
Discount Amount Wrong (CM) =
SUM(Sales[Unit Price Discount Pct]) * SUM(Sales[Extended Amount])
```

Category	Discout Amount (CM)	Discount Amount Wrong (CM)
Bikes	$494,641.03	$25,102,795,191.34
Clothing	$20,964.54	$92,728,708.26
Accessories	$6,688.03	$42,940,331.31
Components	$5,214.74	$23,490,534.87
Total	**$527,508.34**	**$37,828,965,540.70**

Discount Amount Large Sales (CM) =
 VAR LargeSalesTable =
   FILTER(Sales,Sales[Order Quantity] >10)
RETURN
   SUMX(LargeSalesTable,
   Sales[Unit Price Discount Pct] * Sales[Extended Amount])

```
Discount Amount Large Sales (CM) =
 VAR LargeSalesTable =
    FILTER(Sales,Sales[Order Quantity] >10)
RETURN
    SUMX( LargeSalesTable,
    Sales[Unit Price Discount Pct] * Sales[Extended Amount])
```

Category	Discout Amount (CM)	Discount Amount Wrong (CM)	Discount Amount Large Sales (CM)
Bikes	$494,641.03	$25,102,795,191.34	148,798.45
Clothing	$20,964.54	$92,728,708.26	20,964.54
Accessories	$6,688.03	$42,940,331.31	5,314.82
Components	$5,214.74	$23,490,534.87	5,214.74
Total	**$527,508.34**	**$37,828,965,540.70**	**180,292.56**

AVERAGEX() : to find the average delivery days
Average Deivery Date (CM) =
AVERAGEX(Sales, Sales[DeliveryDate]-Sales[OrderDate])

```
Average Deivery Date (CM) =
AVERAGEX(Sales, Sales[DeliveryDate]-Sales[OrderDate])
```

Year	Average Deivery Date (CM)
2017	10.05
2018	9.28
2019	8.21
2020	8.26
Total	**8.47**

```
Average Deivery Date (CM) =
AVERAGEX(Sales, Sales[DeliveryDate]-Sales[OrderDate])
```

Country	Average Deivery Date (CM)
Australia	10.03
Canada	8.27
France	8.17
Germany	8.02
United Kingdom	8.13
United States	8.31
Total	**8.47**

```
Bonus Amount (CM) =
SUMX (
  Sales,
  IF (
    RELATED ( 'Date'[WorkingDays] ) = "WEEKDAY", //Weedend: Friday & Saturday
    Sales[Sales Amount] * 0.01,
    Sales[Sales Amount] * 0.02
  )
)
```

```
Bonus Amount (CM) =
SUMX (
    Sales,
    IF (
        RELATED ( 'Date'[WorkingDays] ) = "WEEKDAY",
        Sales[Sales Amount] * 0.01,
        Sales[Sales Amount] * 0.02
    )
)
```

Year	Average Deivery Date (CM)	Bonus Amount (CM)
2017	10.05	$238,571.13
2018	9.28	$610,337.90
2019	8.21	$857,902.16
2020	8.26	$489,374.29
Total	**8.47**	**$2,196,185.48**

MINX(), MAXX()

Minimum Sales Per Customer (CC) = MINX(RELATEDTABLE(Sales),Sales[Sales Amount])

Maximum Sales Per Customer(CC) = MAXX(RELATEDTABLE(Sales),Sales[Sales Amount])

Customer	Sum of Maximum Sales Per Customer(CC)	Sum of Minimum Sales Per Customer (CC)
Aaron Adams	53.99	3.99
Aaron Alexander	69.99	69.99
Aaron Allen	3,399.99	3,399.99
Aaron Baker	1,700.99	49.99
Aaron Bryant	49.99	4.99
Aaron Butler	9.99	4.99
Aaron Campbell	1,120.49	34.99
Aaron Carter	34.99	4.99
Aaron Chen	34.99	4.99
Aaron Coleman	34.99	4.99
Aaron Collins	3,578.27	34.99
Aaron Diaz	3,578.27	7.95
Aaron Edwards	69.99	24.49
Total	**19,235,706.33**	**2,892,468.78**

AVERAGEX()
Average SAles Per Product (CM) = AVERAGEX('Product',[Total Sales Amount (CM)])

Average Sales Per Product without context Transistion (CM) = AVERAGEX('Product',SUM(Sales[Sales Amount]))

```
Average SAles Per Product (CM) =
AVERAGEX('Product',[Total Sales Amount (CM)])
```

Year	Total Sales Amount (CM)	Average SAles Per Product (CM)
2017	11,928,556.40	$198,809.27
2018	30,516,895.19	$183,836.72
2019	42,895,108.14	$152,110.31
2020	24,468,714.27	$147,401.89
Total	**109,809,274.00**	**$313,740.78**

ture | Formatting | Properties

```
Average Sales Per Product without context Transistion (CM) =
AVERAGEX('Product',SUM(Sales[Sales Amount]))
```

Year	Total Sales Amount (CM)	Average SAles Per Product (CM)	Average SAles Per Product without context Transistion (CM)
2019	42,895,108.14	$152,110.31	42,895,108.14
2018	30,516,895.19	$183,836.72	30,516,895.19
2020	24,468,714.27	$147,401.89	24,468,714.27
2017	11,928,556.40	$198,809.27	11,928,556.40
Total	**109,809,274.00**	**$313,740.78**	**109,809,274.00**

ture | Formatting | Properties

```
Total No of Product Sold (CM) = DISTINCTCOUNT(Sales[ProductKey])
```

Year	Total Sales Amount (CM)	Average SAles Per Product (CM)	Total No of Product Sold (CM)	Average Sal
2019	42,895,108.14	$152,110.31	282	
2018	30,516,895.19	$183,836.72	166	
2020	24,468,714.27	$147,401.89	166	
2017	11,928,556.40	$198,809.27	60	
Total	**109,809,274.00**	**$313,740.78**	**350**	

Average Sales (CM) = AVERAGE(Sales[Sales Amount])

No Of Sales (CM) = COUNTROWS(Sales)

Year	Total Sales Amount (CM)	Average SAles Per Product (CM)	Total No of Product Sold (CM)	Average Sales (CM)	No Of Sales (CM)
2017	11,928,556.40	$198,809.27	60	2,234.23	5339
2018	30,516,895.19	$183,836.72	166	1,572.31	19409
2019	42,895,108.14	$152,110.31	282	772.30	55542
2020	24,468,714.27	$147,401.89	166	597.34	40963
Total	**109,809,274.00**	**$313,740.78**	**350**	**905.62**	**121253**

Advance Filtering in DAX

Calculate:

F_SALES In Australia (CM) = CALCULATE([Total Sales Amount (CM)],SalesTerritory[Country]="Australia")

```
1 F_SALES In Australia (CM) =
2 CALCULATE([Total Sales Amount (CM)],SalesTerritory[Country]="Australia")
```

Category	Total Sales Amount (CM)	F_SALES In Australia (CM)
Accessories	$1,272,059.709999996	$162,638.18
Bikes	$94,620,526.58999181	$10,175,870.58
Clothing	$2,117,613.519999931	$113,175.73
Components	$11,799,074.17999995	$203,651.08
Total	**$109,809,273.9999917**	**$10,655,335.57**

F_Australia Sales % = DIVIDE([F_SALES In Australia (CM)],[Total Sales Amount (CM)],0)

```
F_Australia Sales % =
DIVIDE([F_SALES In Australia (CM)],[Total Sales Amount (CM)],0)
```

Category	Total Sales Amount (CM)	F_SALES In Australia (CM)	F_Australia Sales %
Accessories	$1,272,059.709999996	$162,638.18	12.79%
Bikes	$94,620,526.58999181	$10,175,870.58	10.75%
Clothing	$2,117,613.519999931	$113,175.73	5.34%
Components	$11,799,074.17999995	$203,651.08	1.73%
Total	**$109,809,273.9999917**	**$10,655,335.57**	**9.70%**

```
F_SALES In Australia 2 (CM) =
CALCULATE(
   [Total Sales Amount (CM)],
   FILTER(
       ALL(SalesTerritory[Country]),
      SalesTerritory[Country]= "Australia"
   )
)
```

```
F_SALES In Australia 2 (CM) =
CALCULATE(
    [Total Sales Amount (CM)],
    FILTER(
        ALL(SalesTerritory[Country]),
        SalesTerritory[Country]= "Australia"
    )
)
```

Country	Total Sales Amount (CM)	F_SALES In Australia (CM)	F_Australia Sales %	F_SALES In Australia 2 (CM)
Australia	$10,655,335.57	$10,655,335.57	100.00%	10,655,335.57
Canada	$16,355,770.59999999	$10,655,335.57	65.15%	10,655,335.57
Corporate HQ		$10,655,335.57	0.00%	10,655,335.57
France	$7,251,555.660000009	$10,655,335.57	146.94%	10,655,335.57
Germany	$4,878,300.109999996	$10,655,335.57	218.42%	10,655,335.57
United Kingdom	$7,670,721.130000004	$10,655,335.57	138.91%	10,655,335.57
United States	$62,997,590.92999958	$10,655,335.57	16.91%	10,655,335.57
Total	**$109,809,273.9999996**	**$10,655,335.57**	**9.70%**	**10,655,335.57**

INTERSECT()

KEEPFILTER()

F_Sales in AUS KeepFilter (CM) =
CALCULATE([Total Sales Amount (CM)],
KEEPFILTERS(SalesTerritory[Country] = "Australia"))

```
F_Sales in AUS KeepFilter (CM) =
CALCULATE([Total Sales Amount (CM)],
KEEPFILTERS(SalesTerritory[Country] = "Australia"))
```

Country	Total Sales Amount (CM)	F_Sales in AUS KeepFilter (CM)	F_
Australia	$10,655,335.57	10,655,335.57	
Canada	$16,355,770.59999999		
Corporate HQ			
France	$7,251,555.660000009		
Germany	$4,878,300.109999996		
United Kingdom	$7,670,721.130000004		
United States	$62,997,590.92999958		
Total	**$109,809,273.9999996**	**10,655,335.57**	

Multiple Filter
F_Sales in AUS & CAN =
CALCULATE([Total Sales Amount (CM)],
(SalesTerritory[Country] = "Australia" || SalesTerritory[Country] = "Canada"))

```
F_Sales in AUS & CAN =
CALCULATE([Total Sales Amount (CM)],
(SalesTerritory[Country] = "Australia" || SalesTerritory[Country] = "Canada"))
```

Country	Total Sales Amount (CM)	F_Sales in AUS & CAN	F_Sales in AUS KeepFilter (CM)	F_SALES In Australia
Australia	$10,655,335.57	27,011,106.17	$10,655,335.57	$10,655,3
Canada	$16,355,770.59999999	27,011,106.17		$10,655,3
Corporate HQ		27,011,106.17		$10,655,3
France	$7,251,555.660000009	27,011,106.17		$10,655,3
Germany	$4,878,300.109999996	27,011,106.17		$10,655,3
United Kingdom	$7,670,721.130000004	27,011,106.17		$10,655,3
United States	$62,997,590.92999958	27,011,106.17		$10,655,3
Total	**$109,809,273.9999996**	**27,011,106.17**	**$10,655,335.57**	**$10,655,3**

F_Sales in AUS & CAN KeepFIlter (CM) =
CALCULATE([Total Sales Amount (CM)],
KEEPFILTERS(SalesTerritory[Country] = "Australia" || SalesTerritory[Country] = "Canada"))

```
F_Sales in AUS & CAN KeepFIlter (CM) =
CALCULATE([Total Sales Amount (CM)],
KEEPFILTERS(SalesTerritory[Country] = "Australia" || SalesTerritory[Country] = "Canada"))
```

Country	Total Sales Amount (CM)	F_Sales in AUS & CAN	F_Sales in AUS & CAN KeepFIlter (CM)	F_Sales in AUS Ke
Australia	$10,655,335.57	27,011,106.17	10,655,335.57	$1
Canada	$16,355,770.59999999	27,011,106.17	16,355,770.60	
Corporate HQ		27,011,106.17		
France	$7,251,555.660000009	27,011,106.17		
Germany	$4,878,300.109999996	27,011,106.17		
United Kingdom	$7,670,721.130000004	27,011,106.17		
United States	$62,997,590.92999958	27,011,106.17		
Total	**$109,809,273.9999996**	**27,011,106.17**	**27,011,106.17**	**$1(**

Category	Total Sales Amount (CM)	F_Sales in AUS & CAN	F_Sales in AUS & CAN KeepFIlter (CM)	F_Sales in AUS KeepFilter (CM)	F_SA
Accessories	$1,272,059.709999996	384,143.76	$384,143.76	$162,638.18	
Bikes	$94,620,526.58999181	23,633,553.67	$23,633,553.67	$10,175,870.58	
Clothing	$2,117,613.519999931	545,288.08	$545,288.08	$113,175.73	
Components	$11,799,074.17999995	2,448,120.66	$2,448,120.66	$203,651.08	
Total	**$109,809,273.9999917**	**27,011,106.17**	**$27,011,106.17**	**$10,655,335.57**	

IN Keyword
F_Sales in AUS & CAN & UK(CM) =
CALCULATE([Total Sales Amount (CM)],
SalesTerritory[Country] IN { "Australia", "Canada", "United Kingdom"}
)

```
F_Sales in AUS & CAN & UK(CM) =
CALCULATE([Total Sales Amount (CM)],
SalesTerritory[Country] IN { "Australia","Canada", "United Kingdom"}
)
```

Category	Total Sales Amount (CM)	F_Sales in AUS & CAN & UK(CM)	F_Sales in AUS & CAN	F_Sales in AUS & CAN	F_Sales in AUS & CAN
Accessories	$1,272,059.709999996	503,366.93	384,143.76	384,143.76	
Bikes	$94,620,526.58999181	30,322,143.39	23,633,553.67	23,633,553.67	
Clothing	$2,117,613.519999931	696,356.46	545,288.08	545,288.08	
Components	$11,799,074.17999995	3,159,960.52	2,448,120.66	2,448,120.66	
Total	**$109,809,273.9999917**	**34,681,827.30**	**27,011,106.17**	**27,011,106.17**	**$**

Using Filter with Cslculate
I_Sales when Discount > 500 (CM) =
CALCULATE([Total Sales Amount (CM)],
FILTER(Sales,[Discout Amount (CM)] > 500)
)

```
I_Sales when Discount > 500 (CM) =
CALCULATE([Total Sales Amount (CM)],
FILTER(Sales,[Discout Amount (CM)] > 500)
)
```

Country	Total Sales Amount (CM)	I_Sales when Discount > 500 (CM)
Australia	$10,655,335.57	50,351.56
Canada	$16,355,770.59999999	187,644.71
France	$7,251,555.660000009	172,440.31
Germany	$4,878,300.109999996	80,381.39
United Kingdom	$7,670,721.130000004	106,627.54
Total	**$109,809,273.9999996**	**1,450,720.78**

ALL Sales Amount (CM) = CALCULATE([Total Sales Amount (CM)],ALL(SalesTerritory[Country]))

```
ALL Sales Amount (CM) = CALCULATE([Total Sales Amount (CM)],ALL(SalesTerritory[Country]))
```

Country	Total Sales Amount (CM)	I_Sales when Discount > 500 (CM)	ALL Sales Amount (CM)
Australia	$10,655,335.57	50,351.56	$109,809,274.00
Canada	$16,355,770.59999999	187,644.71	$109,809,274.00
Corporate HQ			$109,809,274.00
France	$7,251,555.660000009	172,440.31	$109,809,274.00
Germany	$4,878,300.109999996	80,381.39	$109,809,274.00
Total	**$109,809,273.9999996**	**1,450,720.78**	**$109,809,274.00**

A_ RM Sales Amount (CM) = CALCULATE([Total Sales Amount (CM)],REMOVEFILTERS(SalesTerritory[Country]))

```
A_ RM Sales Amount (CM) = CALCULATE([Total Sales Amount (CM)],REMOVEFILTERS(SalesTerritory[Country]))
```

Country	Total Sales Amount (CM)	I_Sales when Discount > 500 (CM)	ALL Sales Amount (CM)	A_ RM Sales Amount (CM)
Australia	$10,655,335.57	50,351.56	$109,809,274.00	109,809,274.00
Canada	$16,355,770.59999999	187,644.71	$109,809,274.00	109,809,274.00
Corporate HQ			$109,809,274.00	109,809,274.00
France	$7,251,555.660000009	172,440.31	$109,809,274.00	109,809,274.00
Germany	$4,878,300.109999996	80,381.39	$109,809,274.00	109,809,274.00
Total	**$109,809,273.9999996**	**1,450,720.78**	**$109,809,274.00**	**109,809,274.00**

A_ % of Total Sales (CM) = DIVIDE([Total Sales Amount (CM)],[A_ ALL Sales Amount (CM)],0)

```
1 A_ % of Total Sales (CM) = DIVIDE([Total Sales Amount (CM)],[A_ ALL Sales Amount (CM)],0)
```

Country	Total Sales Amount (CM)	I_Sales when Discount > 500 (CM)	A_ ALL Sales Amount (CM)	A_ RM Sales Amount (CM)	A_ % of Total Sales (CM)
Australia	$10,655,335.57	50,351.56	$109,809,274.00	109,809,274.00	9.70%
Canada	$16,355,770.59999999	187,644.71	$109,809,274.00	109,809,274.00	14.89%
Corporate HQ			$109,809,274.00	109,809,274.00	
France	$7,251,555.660000009	172,440.31	$109,809,274.00	109,809,274.00	6.60%
Germany	$4,878,300.109999996	80,381.39	$109,809,274.00	109,809,274.00	4.44%
Total	**$109,809,273.9999996**	**1,450,720.78**	**$109,809,274.00**	**109,809,274.00**	**100.00%**

Slider to test ALLSELECTED

Country	Total Sales Amount (CM)	A_ % of Total Sales (CM)
France	$7,251,555.660000009	36.62%
Germany	$4,878,300.109999996	24.64%
United Kingdom	$7,670,721.130000004	38.74%
Total	**$19,800,576.90000001**	**100.00%**

Group
- ○ Corporate HQ
- ● Europe
- ○ North America
- ○ Pacific

_A ALLselected Country Sales (CM)= CALCULATE([Total Sales Amount (CM)], ALLSELECTED(SalesTerritory))

_A % of Total Sales using ALLSELECTED (CM) = DIVIDE([Total Sales Amount (CM)],[_A ALLselected Country Sales (CM)],0)

```
1 _A % of Total Sales using ALLSELECTED (CM) =
2 DIVIDE([Total Sales Amount (CM)],[_A ALLselected Country Sales (CM)],0)
```

Country	Total Sales Amount (CM)	A_ % of Total Sales (CM)	_A ALLselected Country Sales (CM)	_A % of Total Sales using ALLSELECTED (CM)
Australia	$10,655,335.57	9.70%	$109,809,274.00	9.70%
Canada	$16,355,770.59999999	14.89%	$109,809,274.00	14.89%
Corporate HQ			$109,809,274.00	
France	$7,251,555.660000009	6.60%	$109,809,274.00	6.60%
Germany	$4,878,300.109999996	4.44%	$109,809,274.00	4.44%
United Kingdom	$7,670,721.130000004	6.99%	$109,809,274.00	6.99%
United States	$62,997,590.92999958	57.37%	$109,809,274.00	57.37%
Total	**$109,809,273.9999996**	**100.00%**	**$109,809,274.00**	**100.00%**

Group
- ○ Corporate HQ
- ○ Europe
- ○ North America
- ○ Pacific

```
1 _A % of Total Sales using ALLSELECTED (CM) =
2 DIVIDE([Total Sales Amount (CM)],[_A ALLselected Country Sales (CM)],0)
```

Country	Total Sales Amount (CM)	A_ % of Total Sales (CM)	_A ALLselected Country Sales (CM)	_A % of Total Sales using ALLSELECTED (CM)
France	$7,251,555.660000009	36.62%	$19,800,576.90	36.62%
Germany	$4,878,300.109999996	24.64%	$19,800,576.90	24.64%
United Kingdom	$7,670,721.130000004	38.74%	$19,800,576.90	38.74%
Total	**$19,800,576.90000001**	**100.00%**	**$19,800,576.90**	**100.00%**

Group
- ○ Corporate HQ
- ● Europe
- ○ North America
- ○ Pacific

Time Intelligence

```
TI_Custom YTD (CM) =
 CALCULATE(
   SUM(Sales[Sales Amount]),
   'Date'[Date] >= DATE(2018,1,1)
   && 'Date'[Date] <= DATE(2018,5,18)
 )
```

```
TI_Custom YTD (CM) =
 CALCULATE(
    SUM(Sales[Sales Amount]),
    'Date'[Date] >= DATE(2018,1,1)
    && 'Date'[Date] <= DATE(2018,5,18)
 )

```

Year	Total Sales Amount (CM)	TI_Custom YTD (CM)
2017	**$11,928,556.40**	**$9,122,073.48**
Jan		$9,122,073.48
Feb		$9,122,073.48
Mar		$9,122,073.48
Apr		$9,122,073.48
May		$9,122,073.48
Jun		$9,122,073.48
Jul	$1,423,357.36	$9,122,073.48
Aug	$2,057,902.54	$9,122,073.48
Sep	$2,523,947.76	$9,122,073.48
Oct	$561,681.51	$9,122,073.48
Nov	$4,764,920.63	$9,122,073.48
Dec	$596,746.60	$9,122,073.48
2018	**$30,516,895.19**	**$9,122,073.48**
2019	**$42,895,108.14**	**$9,122,073.48**
2020	**$24,468,714.27**	**$9,122,073.48**
Total	**$109,809,274.00**	**$9,122,073.48**

DATESBETWEEN()

```
TI_Custom YTD DB (CM) =
 CALCULATE(
   SUM(Sales[Sales Amount]),
   DATESBETWEEN('Date'[Date],DATE(2018,1,1),DATE(2018,5,18)
  )
)
```

```
TI_Custom YTD DB (CM) =
 CALCULATE(
    SUM(Sales[Sales Amount]),
    DATESBETWEEN('Date'[Date],DATE(2018,1,1),DATE(2018,5,18)
   )
)
```

Year	Total Sales Amount (CM)	TI_Custom YTD (CM)	TI_Custom YTD DB (CM)
2017	**$11,928,556.40**	**$9,122,073.48**	**9,122,073.48**
Jan		$9,122,073.48	9,122,073.48
Feb		$9,122,073.48	9,122,073.48
Mar		$9,122,073.48	9,122,073.48
Apr		$9,122,073.48	9,122,073.48
May		$9,122,073.48	9,122,073.48
Jun		$9,122,073.48	9,122,073.48
Jul	$1,423,357.36	$9,122,073.48	9,122,073.48
Aug	$2,057,902.54	$9,122,073.48	9,122,073.48
Sep	$2,523,947.76	$9,122,073.48	9,122,073.48
Oct	$561,681.51	$9,122,073.48	9,122,073.48
Nov	$4,764,920.63	$9,122,073.48	9,122,073.48
Dec	$596,746.60	$9,122,073.48	9,122,073.48
2018	**$30,516,895.19**	**$9,122,073.48**	**9,122,073.48**
2019	**$42,895,108.14**	**$9,122,073.48**	**9,122,073.48**
2020	**$24,468,714.27**	**$9,122,073.48**	**9,122,073.48**
Total	**$109,809,274.00**	**$9,122,073.48**	**9,122,073.48**

```
TI_ Custom YTD using Variable =
VAR LastDateOfSales = MAX('Date'[Date])
RETURN
CALCULATE(
   SUM(Sales[Sales Amount]),
   DATESBETWEEN('Date'[Date],DATE(YEAR(LastDateOfSales),1,1),
   LastDateOfSales)
)
```

```
TI_ Custom YTD using Variable =
VAR LastDateOfSales = MAX('Date'[Date])
RETURN
CALCULATE(
    SUM(Sales[Sales Amount]),
    DATESBETWEEN('Date'[Date],DATE(YEAR(LastDateOfSales),1,1),
    LastDateOfSales)
)
```

Year	Total Sales Amount (CM)	TI_Custom YTD DB (CM)	TI_ Custom YTD using Variable
2017	**$11,928,556.40**	**9,122,073.48**	**$11,928,556.40**
Jan		9,122,073.48	
Feb		9,122,073.48	
Mar		9,122,073.48	
Apr		9,122,073.48	
May		9,122,073.48	
Jun		9,122,073.48	
Jul	$1,423,357.36	9,122,073.48	$1,423,357.36
Aug	$2,057,902.54	9,122,073.48	$3,481,259.90
Sep	$2,523,947.76	9,122,073.48	$6,005,207.66
Oct	$561,681.51	9,122,073.48	$6,566,889.17
Nov	$4,764,920.63	9,122,073.48	$11,331,809.80
Dec	$596,746.60	9,122,073.48	$11,928,556.40
2018	**$30,516,895.19**	**9,122,073.48**	**$30,516,895.19**
2019	**$42,895,108.14**	**9,122,073.48**	**$42,895,108.14**
2020	**$24,468,714.27**	**9,122,073.48**	**$24,468,714.27**
Total	**$109,809,274.00**	**9,122,073.48**	**$24,468,714.27**

Using DATESYTD()

TI_Sales Amount YTD using DATESYD (CM) = CALCULATE([Total Sales Amount (CM)], DATESYTD('Date'[Date]))

Structure | Formatting | Properties | Calculations

```
1 TI_Sales Amount YTD using DATESYD (CM) =
2 CALCULATE([Total Sales Amount (CM)], DATESYTD('Date'[Date]))
```

Year	Total Sales Amount (CM)	TI_Custom YTD DB (CM)	TI_ Custom YTD using Variable	TI_Sales Amount YTD using DATESYD (CM)
2017	**$11,928,556.40**	**9,122,073.48**	**$11,928,556.40**	**$11,928,556.40**
Jan		9,122,073.48		
Feb		9,122,073.48		
Mar		9,122,073.48		
Apr		9,122,073.48		
May		9,122,073.48		
Jun		9,122,073.48		
Jul	$1,423,357.36	9,122,073.48	$1,423,357.36	$1,423,357.36
Aug	$2,057,902.54	9,122,073.48	$3,481,259.90	$3,481,259.90
Sep	$2,523,947.76	9,122,073.48	$6,005,207.66	$6,005,207.66
Oct	$561,681.51	9,122,073.48	$6,566,889.17	$6,566,889.17
Nov	$4,764,920.63	9,122,073.48	$11,331,809.80	$11,331,809.80
Dec	$596,746.60	9,122,073.48	$11,928,556.40	$11,928,556.40
2020	**$24,468,714.27**	**9,122,073.48**	**$24,468,714.27**	**$24,468,714.27**
2018	**$30,516,895.19**	**9,122,073.48**	**$30,516,895.19**	**$30,516,895.19**
2019	**$42,895,108.14**	**9,122,073.48**	**$42,895,108.14**	**$42,895,108.14**
Total	**$109,809,274.00**	**9,122,073.48**	**$24,468,714.27**	**$24,468,714.27**

Using TOTALYTD()

TI_Sales Amount YTD using TOTALYTD (CM) =
TOTALYTD([Total Sales Amount (CM)],'Date'[Date])

Structure | Formatting | Properties | Calculations

```
1 TI_Sales Amount YTD using TOTALYTD (CM) =
2 TOTALYTD([Total Sales Amount (CM)],'Date'[Date])
```

Year	Total Sales Amount (CM)	TI_Custom YTD DB (CM)	TI_ Custom YTD using Variable	TI_Sales Amount YTD using DATESYD (CM)	TI_Sales Amount YTD using TOTALYTD (CM)
2017	**$11,928,556.40**	**9,122,073.48**	**$11,928,556.40**	**$11,928,556.40**	**$11,928,556.40**
Jan		9,122,073.48			
Feb		9,122,073.48			
Mar		9,122,073.48			
Apr		9,122,073.48			
May		9,122,073.48			
Jun		9,122,073.48			
Jul	$1,423,357.36	9,122,073.48	$1,423,357.36	$1,423,357.36	$1,423,357.36
Aug	$2,057,902.54	9,122,073.48	$3,481,259.90	$3,481,259.90	$3,481,259.90
Sep	$2,523,947.76	9,122,073.48	$6,005,207.66	$6,005,207.66	$6,005,207.66
Oct	$561,681.51	9,122,073.48	$6,566,889.17	$6,566,889.17	$6,566,889.17
Nov	$4,764,920.63	9,122,073.48	$11,331,809.80	$11,331,809.80	$11,331,809.80
Dec	$596,746.60	9,122,073.48	$11,928,556.40	$11,928,556.40	$11,928,556.40
2020	**$24,468,714.27**	**9,122,073.48**	**$24,468,714.27**	**$24,468,714.27**	**$24,468,714.27**
2018	**$30,516,895.19**	**9,122,073.48**	**$30,516,895.19**	**$30,516,895.19**	**$30,516,895.19**
2019	**$42,895,108.14**	**9,122,073.48**	**$42,895,108.14**	**$42,895,108.14**	**$42,895,108.14**
Total	**$109,809,274.00**	**9,122,073.48**	**$24,468,714.27**	**$24,468,714.27**	**$24,468,714.27**

Handling Fiscal Year

TI_Sales Amount YTD for Fiscal Year (CM) = TOTALYTD([Total Sales Amount (CM)],'Date'[Date],"06-30")

Structure Formatting Properties Calculations

```
1 TI_Sales Amount YTD for Fiscal Year (CM) =
2 TOTALYTD([Total Sales Amount (CM)],'Date'[Date],"06-30")
```

Year	Total Sales Amount (CM)	TI_Custom YTD DB (CM)	TI_ Custom YTD using Variable	TI_Sales Amount YTD using DATESYD (CM)	TI_Sales Amount YTD using TOTALYTD (CM)	TI_Sales Amount YTD for Fiscal Year (CM)
2017	**$11,928,556.40**	**9,122,073.48**	**$11,928,556.40**	**$11,928,556.40**	**$11,928,556.40**	**$11,928,556.40**
2020	**$24,468,714.27**	**9,122,073.48**	**$24,468,714.27**	**$24,468,714.27**	**$24,468,714.27**	
2018	**$30,516,895.19**	**9,122,073.48**	**$30,516,895.19**	**$30,516,895.19**	**$30,516,895.19**	**$18,584,558.46**
Jan	$1,327,674.78	9,122,073.48	$1,327,674.78	$1,327,674.78	$1,327,674.78	$13,256,231.18
Feb	$3,936,463.67	9,122,073.48	$5,264,138.45	$5,264,138.45	$5,264,138.45	$17,192,694.85
Mar	$700,873.23	9,122,073.48	$5,965,011.68	$5,965,011.68	$5,965,011.68	$17,893,568.08
Apr	$1,519,275.52	9,122,073.48	$7,484,287.20	$7,484,287.20	$7,484,287.20	$19,412,843.60
May	$2,960,378.17	9,122,073.48	$10,444,665.37	$10,444,665.37	$10,444,665.37	$22,373,221.77
Jun	$1,487,671.36	9,122,073.48	$11,932,336.73	$11,932,336.73	$11,932,336.73	$23,860,893.13
Jul	$2,939,691.10	9,122,073.48	$14,872,027.83	$14,872,027.83	$14,872,027.83	$2,939,691.10
Aug	$3,964,801.97	9,122,073.48	$18,836,829.80	$18,836,829.80	$18,836,829.80	$6,904,493.07
Sep	$3,287,606.21	9,122,073.48	$22,124,436.01	$22,124,436.01	$22,124,436.01	$10,192,099.28
Oct	$2,157,287.80	9,122,073.48	$24,281,723.81	$24,281,723.81	$24,281,723.81	$12,349,387.08
Nov	$3,611,092.66	9,122,073.48	$27,892,816.47	$27,892,816.47	$27,892,816.47	$15,960,479.74
Dec	$2,624,078.72	9,122,073.48	$30,516,895.19	$30,516,895.19	$30,516,895.19	$18,584,558.46
2019	**$42,895,108.14**	**9,122,073.48**	**$42,895,108.14**	**$42,895,108.14**	**$42,895,108.14**	**$27,409,553.93**
Total	**$109,809,274.00**	**9,122,073.48**	**$24,468,714.27**	**$24,468,714.27**	**$24,468,714.27**	

SAME PRICE LAST YEAR

DATEADD()

TI_Sale SPLY using DATEADD (CM) = CALCULATE([Total Sales Amount (CM)],DATEADD('Date'[Date],-1,YEAR)

```
1 TI_Sale SPLY using DATEADD (CM) =
2 CALCULATE([Total Sales Amount (CM)],DATEADD('Date'[Date],-1,YEAR))
```

Year	Total Sales Amount (CM)	TI_Sale SPLY using DATEADD (CM)
2017	$11,928,556.40	
2018	$30,516,895.19	$11,928,556.40
2019	$42,895,108.14	$30,516,895.19
2020	$24,468,714.27	$42,895,108.14
Total	**$109,809,274.00**	**$85,340,559.73**

Using SAMEPERIODLASTYEAR()

```
TI_SPLY using SAMEPERIODLASTYEAR (CM) =
CALCULATE([Total Sales Amount (CM)],SAMEPERIODLASTYEAR('Date'[Date]))
```

Year	Total Sales Amount (CM)	TI_Sale SPLY using DATEADD (CM)	TI_SPLY using SAMEPERIODLASTYEAR (CM)
2017	**$11,928,556.40**		
2018	**$30,516,895.19**	**$11,928,556.40**	**$11,928,556.40**
Jan	$1,327,674.78		
Feb	$3,936,463.67		
Mar	$700,873.23		
Apr	$1,519,275.52		
May	$2,960,378.17		
Jun	$1,487,671.36		
Jul	$2,939,691.10	$1,423,357.36	$1,423,357.36
Aug	$3,964,801.97	$2,057,902.54	$2,057,902.54
Sep	$3,287,606.21	$2,523,947.76	$2,523,947.76
Oct	$2,157,287.80	$561,681.51	$561,681.51
Nov	$3,611,092.66	$4,764,920.63	$4,764,920.63
Dec	$2,624,078.72	$596,746.60	$596,746.60
2019	**$42,895,108.14**	**$30,516,895.19**	**$30,516,895.19**
2020	**$24,468,714.27**	**$42,895,108.14**	**$42,895,108.14**
Total	**$109,809,274.00**	**$85,340,559.73**	**$85,340,559.73**

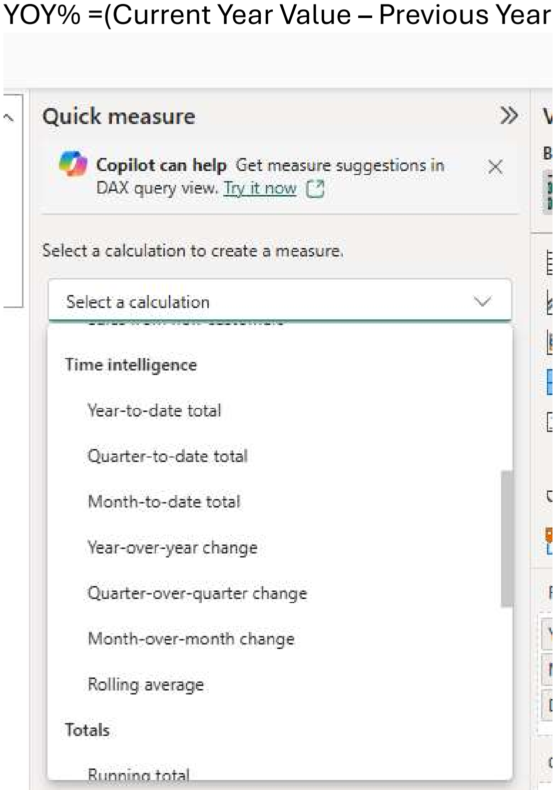
Finding YOY %

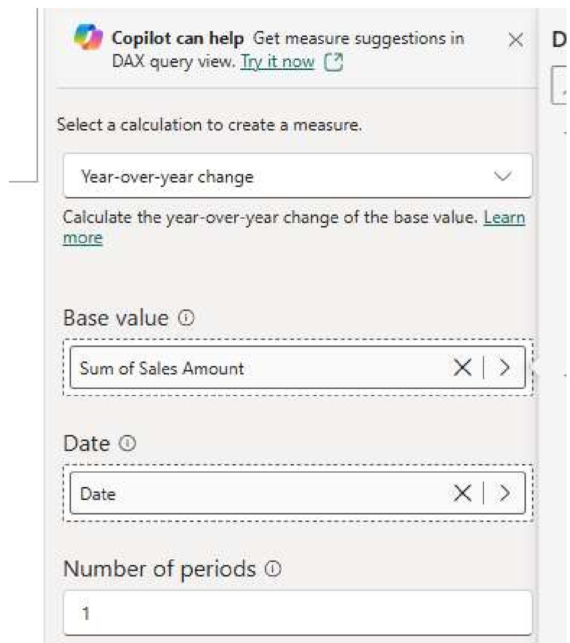
```
TI_ Custom YOY% (CM)=
VAR CurrentSales =[Total Sales Amount (CM)]
VAR PreviousSales = [TI_SPLY using SAMEPERIODLASTYEAR (CM)]
VAR DeltaSales = CurrentSales-PreviousSales
VAR Result =
  IF(NOT ISBLANK(CurrentSales),DIVIDE(DeltaSales,PreviousSales))
RETURN
  Result
```

```
TI_ Custom YOY% (CM) =
VAR CurrentSales =[Total Sales Amount (CM)]
VAR PreviousSales = [TI_SPLY using SAMEPERIODLASTYEAR (CM)]
VAR DeltaSales = CurrentSales-PreviousSales
VAR Result =
    IF(NOT ISBLANK(CurrentSales),DIVIDE(DeltaSales,PreviousSales))
RETURN
    Result
```

Year	Total Sales Amount (CM)	TI_ Custom YOY% (CM)
2017	$11,928,556.40	
2018	$30,516,895.19	155.83%
2019	$42,895,108.14	40.56%
2020	$24,468,714.27	-42.96%
Total	**$109,809,274.00**	**28.67%**

YOY% =(Current Year Value – Previous Year Value)/Previous Year value





Quick Measure

```
Sum of Sales Amount YoY% =
VAR __PREV_YEAR = CALCULATE(SUM('Sales'[Sales Amount]), DATEADD('Date'[Date], -1, YEAR))
RETURN
    DIVIDE(SUM('Sales'[Sales Amount]) - __PREV_YEAR, __PREV_YEAR)
```

Year	Total Sales Amount (CM)	TI_ Custom YOY% (CM)	Sum of Sales Amount YoY%
2017	$11,928,556.40		
2018	$30,516,895.19	155.83%	155.83%
2019	$42,895,108.14	40.56%	40.56%
2020	$24,468,714.27	-42.96%	-42.96%
Total	**$109,809,274.00**	**28.67%**	**28.67%**

Using PARALLELPERIOD()

TI_Sales PPLY (CM)=
CALCULATE([Total Sales Amount (CM)],
PARALLELPERIOD('Date'[Date],-1,YEAR)

```
TI_Sales PPLy (CM) =
CALCULATE([Total Sales Amount (CM)],
PARALLELPERIOD('Date'[Date],-1,YEAR))
```

Year	Total Sales Amount (CM)	TI_Sales PPLy (CM)
2017	$11,928,556.40	
2018	$30,516,895.19	$11,928,556.40
2019	$42,895,108.14	$30,516,895.19
2020	$24,468,714.27	$42,895,108.14
Total	**$109,809,274.00**	**$85,340,559.73**

TI_ % Sales compared to last Year (CM) =
DIVIDE([TI_Sales Amount YTD using TOTALYTD (CM)],[TI_Sales PPLy (CM)])

```
TI_ % Sales compared to last Year (CM) =
DIVIDE([TI_Sales Amount YTD using TOTALYTD (CM)],[TI_Sales PPLy (CM)])
```

Year	Total Sales Amount (CM)	TI_Sales PPLy (CM)	TI_ % Sales compared to last Year (CM)
2017	$11,928,556.40		
2018	$30,516,895.19	$11,928,556.40	255.83%
2019	$42,895,108.14	$30,516,895.19	140.56%
2020	$24,468,714.27	$42,895,108.14	57.04%
Total	**$109,809,274.00**	**$85,340,559.73**	**28.67%**

FINDING MOVING TOTAL

Using DATESINPERIOD()

Moving Total for 12 Months (CM) =
CALCULATE(
  [Total Sales Amount (CM)],
  DATESINPERIOD('Date'[Date],MAX('Date'[Date]),-1,YEAR)

```
Moving Total for 12 Months (CM) =
CALCULATE(
    [Total Sales Amount (CM)],
    DATESINPERIOD('Date'[Date],MAX('Date'[Date]),-1,YEAR))
```

Year	Total Sales Amount (CM)	Moving Total for 12 Months (CM)
2017	$11,928,556.40	$11,928,556.40
2018	$30,516,895.19	$30,516,895.19
2019	$42,895,108.14	$42,895,108.14
2020	$24,468,714.27	$24,468,714.27
Total	**$109,809,274.00**	**$24,468,714.27**

Running Total

Running Total (CM)=
VAR LastVisibleDate =
  MAX('Date'[Date])
RETURN
  CALCULATE(
    [Total Sales Amount (CM)],

FILTER(ALL('Date'),'Date'[Date] <= LastVisibleDate)

```
Running Total (CM) =
VAR LastVisibleDate =
    MAX('Date'[Date])
RETURN
    CALCULATE(
        [Total Sales Amount (CM)],
        FILTER(ALL('Date'),'Date'[Date] <= LastVisibleDate)
        )
```

Year	Total Sales Amount (CM)	Moving Total for 12 Months (CM)	Running Total (CM)
2017	$11,928,556.40	$11,928,556.40	$11,928,556.40
2018	$30,516,895.19	$30,516,895.19	$42,445,451.59
2019	$42,895,108.14	$42,895,108.14	$85,340,559.73
2020	$24,468,714.27	$24,468,714.27	$109,809,274.00
Total	**$109,809,274.00**	**$24,468,714.27**	**$109,809,274.00**

Conditionals in DAX

IF()

CC :

Product Profit (CC) = 'Product'[List Price] - 'Product'[Standard Cost]

Product Profit % (CC) = 'Product'[Product Profit (CC)]/ 'Product'[List Price]

Product Group (CC) =
IF('Product'[Product Profit % (CC)]>0.5, "HIGH",
  IF('Product'[Product Profit % (CC)]>0.3,"Medium","Low")
)

	Subcategory	Category	SKU	Product Profit (CC)	Product Profit % (CC)	Product Group (CC)
	Mountain Bikes	Bikes	BK-M18S-52	$256.77	45.45%	Medium
	Mountain Bikes	Bikes	BK-M68S-42	$1,054.37	45.45%	Medium
	Mountain Bikes	Bikes	BK-M68S-38	$1,054.37	45.45%	Medium
	Mountain Bikes	Bikes	BK-M68S-46	$1,054.37	45.45%	Medium
	Mountain Bikes	Bikes	BK-M68S-38	$953.56	46.03%	Medium
	Mountain Bikes	Bikes	BK-M68S-42	$953.56	46.03%	Medium
	Mountain Bikes	Bikes	BK-M68S-46	$953.56	46.03%	Medium
e	Mountain Frames	Components	FR-M94S-38	$579.54	46.72%	Medium
e	Mountain Frames	Components	FR-M94S-42	$579.54	46.72%	Medium
e	Mountain Frames	Components	FR-M94S-46	$579.54	46.72%	Medium
e	Mountain Frames	Components	FR-M94S-42	$580.48	48.20%	Medium
e	Mountain Frames	Components	FR-M94S-46	$580.48	48.20%	Medium
e	Mountain Frames	Components	FR-M94S-38	$580.48	48.20%	Medium
e	Mountain Frames	Components	FR-M94S-44	$657.69	48.20%	Medium
e	Mountain Frames	Components	FR-M94S-52	$657.69	48.20%	Medium
	Chains	Components	CH-0234	$11.25	55.60%	HIGH
ic	Saddles	Components	SE-T762	$21.76	55.60%	HIGH
'S	Saddles	Components	SE-M798	$21.76	55.60%	HIGH
il	Saddles	Components	SE-R908	$21.76	55.60%	HIGH
s	Handlebars	Components	HB-R504	$22.51	55.60%	HIGH
e	Handlebars	Components	HB-M243	$22.51	55.60%	HIGH

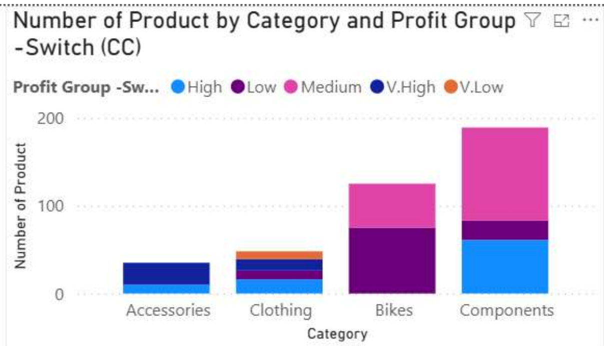
Using Switch()

```
Profit Group -Switch (CC) =
SWITCH(
   TRUE(),
   'Product'[Product Profit % (CC)] > 0.6, "V.High",
   'Product'[Product Profit % (CC)] > 0.5, "High",
   'Product'[Product Profit % (CC)] > 0.4, "Medium",
   'Product'[Product Profit % (CC)] > 0.3, "Low",
   "V.Low"
)
```

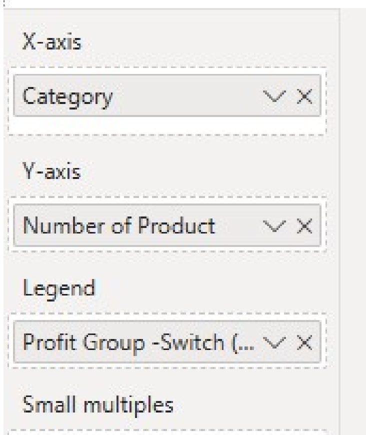
	Subcategory	Category	SKU	Product Profit (CC)	Product Profit % (CC)	Product Group (CC)	Profit Group -Switch (CC)
ame	Mountain Frames	Components	FR-M94S-42	$580.48	48.20%	Medium	V.Low
ame	Mountain Frames	Components	FR-M94S-46	$580.48	48.20%	Medium	V.Low
ame	Mountain Frames	Components	FR-M94S-38	$580.48	48.20%	Medium	V.Low
ame	Mountain Frames	Components	FR-M94S-44	$657.69	48.20%	Medium	V.Low
ame	Mountain Frames	Components	FR-M94S-52	$657.69	48.20%	Medium	V.Low
	Chains	Components	CH-0234	$11.25	55.60%	HIGH	High
t/Sac	Saddles	Components	SE-T762	$21.76	55.60%	HIGH	High
eat/S	Saddles	Components	SE-M798	$21.76	55.60%	HIGH	High
addl	Saddles	Components	SE-R908	$21.76	55.60%	HIGH	High
bars	Handlebars	Components	HB-R504	$22.51	55.60%	HIGH	High
ndle	Handlebars	Components	HB-M243	$22.51	55.60%	HIGH	High
dal	Pedals	Components	PD-M282	$22.51	55.60%	HIGH	High

```
Profit Group-SORT -Switch (CC) =
SWITCH(
   TRUE(),
   'Product'[Product Profit % (CC)] > 0.6, 1,
   'Product'[Product Profit % (CC)] > 0.5, 2,
   'Product'[Product Profit % (CC)] > 0.4, 3,
   'Product'[Product Profit % (CC)] > 0.3, 4,
```

5
)



Category	High	Low	Medium	V.High	V.Low	Total
Accessories	10			25		**35**
Bikes		75	50			**125**
Clothing	16	10		13	9	**48**
Components	61	22	106			**189**
Total	**87**	**107**	**156**	**38**	**9**	**397**



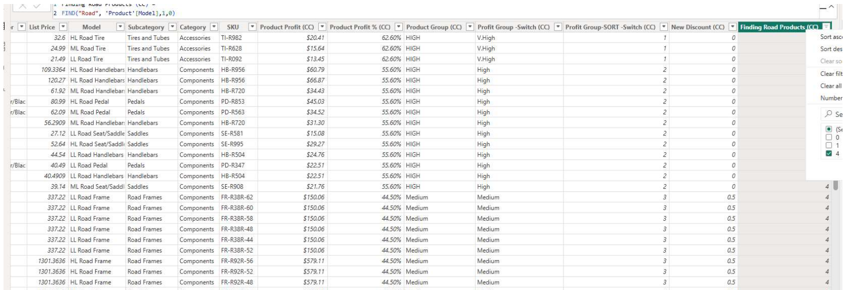
Switching Between Cases using Switch

```
New Discount (CC) =
SWITCH(
   TRUE(),
   'Product'[Color] = "Red", 0.50,
   AND('Product'[Profit Group -Switch (CC)]="Low",'Product'[Color]="Yellow"),0.40, 0
)
```

Category	0.00	0.40	0.50	Total
Accessories	32		3	**35**
Bikes	75	23	27	**125**
Clothing	48			**48**
Components	151	5	33	**189**
Total	**306**	**28**	**63**	**397**

Finding Your Text

Finding Road Products (CC) = FIND("Road", 'Product'[Model],1,0)



```
1 Finding Road Products (CC) =
2 FIND("Road", 'Product'[Model],1,0)
```

r	List Price	Model	Subcategory	Category	SKU	Product Profit (CC)	Product Profit % (CC)	Product Group (CC)	Profit Group -Switch (CC)	Profit Group-SORT -Switch (CC)	New Discount (CC)	Finding Road Products (CC)
	32.6	HL Road Tire	Tires and Tubes	Accessories	TI-R982	$20.41	62.60%	HIGH	V.High	1	0	
	24.99	ML Road Tire	Tires and Tubes	Accessories	TI-R628	$15.64	62.60%	HIGH	V.High	1	0	
	21.49	LL Road Tire	Tires and Tubes	Accessories	TI-R092	$13.45	62.60%	HIGH	V.High	1	0	
	109.3364	HL Road Handlebars	Handlebars	Components	HB-R956	$60.79	55.60%	HIGH	High	2	0	
	120.27	HL Road Handlebars	Handlebars	Components	HB-R956	$66.87	55.60%	HIGH	High	2	0	
	61.92	ML Road Handlebars	Handlebars	Components	HB-R720	$34.43	55.60%	HIGH	High	2	0	
r/Blac	80.99	HL Road Pedal	Pedals	Components	PD-R853	$45.03	55.60%	HIGH	High	2	0	
r/Blac	62.09	ML Road Pedal	Pedals	Components	PD-R563	$34.52	55.60%	HIGH	High	2	0	
	56.2909	ML Road Handlebars	Handlebars	Components	HB-R720	$31.30	55.60%	HIGH	High	2	0	
	27.12	LL Road Seat/Saddle	Saddles	Components	SE-R581	$15.08	55.60%	HIGH	High	2	0	
	52.64	HL Road Seat/Saddle	Saddles	Components	SE-R995	$29.27	55.60%	HIGH	High	2	0	
	44.54	LL Road Handlebars	Handlebars	Components	HB-R504	$24.76	55.60%	HIGH	High	2	0	
r/Blac	40.49	LL Road Pedal	Pedals	Components	PD-R347	$22.51	55.60%	HIGH	High	2	0	
	40.4909	LL Road Handlebars	Handlebars	Components	HB-R504	$22.51	55.60%	HIGH	High	2	0	
	39.14	ML Road Seat/Saddl	Saddles	Components	SE-R908	$21.76	55.60%	HIGH	High	2	0	4
	337.22	LL Road Frame	Road Frames	Components	FR-R38R-62	$150.06	44.50%	Medium	Medium	3	0.5	4
	337.22	LL Road Frame	Road Frames	Components	FR-R38R-60	$150.06	44.50%	Medium	Medium	3	0.5	4
	337.22	LL Road Frame	Road Frames	Components	FR-R38R-58	$150.06	44.50%	Medium	Medium	3	0.5	4
	337.22	LL Road Frame	Road Frames	Components	FR-R38R-48	$150.06	44.50%	Medium	Medium	3	0.5	4
	337.22	LL Road Frame	Road Frames	Components	FR-R38R-44	$150.06	44.50%	Medium	Medium	3	0.5	4
	337.22	LL Road Frame	Road Frames	Components	FR-R38R-52	$150.06	44.50%	Medium	Medium	3	0.5	4
	1301.3636	HL Road Frame	Road Frames	Components	FR-R92R-56	$579.11	44.50%	Medium	Medium	3	0.5	4
	1301.3636	HL Road Frame	Road Frames	Components	FR-R92R-52	$579.11	44.50%	Medium	Medium	3	0.5	4
	1301.3636	HL Road Frame	Road Frames	Components	FR-R92R-48	$579.11	44.50%	Medium	Medium	3	0.5	4

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Other Function

Using CONCATENATEX()

```
X MAX Daily Sales =
MAXX('Date' ,[Total Sales Amount (CM)])

X Date of MAX Sales (CM =
VAR MaxDailySales = [X MAX Daily Sales]
VAR DateWithMaxSales =
   FILTER('Date',[Total Sales Amount (CM)]= MaxDailySales)
VAR Result =
   CONCATENATEX( DateWithMaxSales, FORMAT('Date'[Date], "mm/dd/yy"),", " )
RETURN
   IF(NOT ISBLANK(MaxDailySales), Result, BLANK())
```

Year	X MAX Daily Sales	X Date of MAX Sales (CM
2017	**276783.36**	**11/26/17**
Q3	**220081.82**	**09/12/17**
Jul	116318.97	07/29/17
Aug	181994.51	08/13/17
Sep	220081.82	09/12/17
Q4	**276783.36**	**11/26/17**
2018	**374697.42**	**09/14/18**
2019	**456533.50**	**07/18/19**
2020	**569749.73**	**06/06/20**
Total	**569749.73**	**06/06/20**

HASONEVALUE() & CONCATENATEX()

```
HX Product Keys of Duplicate Products (CM) =
IF (
   HASONEVALUE('Product'[Product]),
   CONCATENATEX('Product','Product'[ProductKey],","))

HX Product List Price of Duplicate Products (CM) =
IF(
   HASONEVALUE('Product'[Product]),
   CONCATENATEX('Product',ROUND('Product'[List Price],1),", "))
```

Product	First SKU	Total No of Product (CM)	HX Product Keys of Duplicate Products (CM)	HX Product List Price of Duplicate Products (CM)
All-Purpose Bike Stand	ST-1401	1	486	159
AWC Logo Cap	CA-1098	3	225,223,224	9, 8.6, 8.6
Bike Wash - Dissolver	CL-9009	1	484	8
Cable Lock	LO-C100	1	447	25
Chain	CH-0234	1	559	20.2
Classic Vest, L	VE-C304-L	1	473	63.5
Classic Vest, M	VE-C304-M	1	472	63.5
Classic Vest, S	VE-C304-S	1	471	63.5
Fender Set - Mountain	FE-6654	1	485	22
Front Brakes	FB-9873	1	555	106.5
Front Derailleur	FD-2342	1	552	91.5
Full-Finger Gloves, L	GL-F110-L	1	470	38
Full-Finger Gloves, M	GL-F110-M	1	469	38
Full-Finger Gloves, S	GL-F110-S	1	468	38
Half-Finger Gloves, L	GL-H102-L	2	466,467	23.5, 24.5
Half-Finger Gloves, M	GL-H102-M	2	464,465	23.5, 24.5
Total	**BB-7421**	**397**		

Using ISINSCOPFE()
OF_ ALL Product Sales (CM) =
CALCULATE([Total Sales Amount (CM)],REMOVEFILTERS('Product')

OF_ % Product Sales (CM) = DIVIDE([Total Sales Amount (CM)],[A_ ALL Sales Amount (CM)],0)

Expected Result:

Category	Total Sales Amount (CM)	% Product Sales (CM)	% ProductSales Revised (CM)
⊟ **Accessories**	**$1,272,059.71**	**1.16 %**	**1.16 %**
Bike Racks	$237,096.21	0.22 %	18.64 %
Bike Stands	$39,591.00	0.04 %	3.11 %
Bottles and Cages	$64,274.95	0.06 %	5.05 %
Cleaners	$18,407.03	0.02 %	1.45 %
Fenders	$46,619.58	0.04 %	3.66 %
Helmets	$484,049.97	0.44 %	38.05 %
Hydration Packs	$105,826.62	0.10 %	8.32 %
Locks	$16,225.22	0.01 %	1.28 %
Pumps	$13,514.71	0.01 %	1.06 %
Tires and Tubes	$246,454.42	0.22 %	19.37 %
⊞ **Bikes**	**$94,620,526.59**	**86.17 %**	**86.17 %**
⊞ **Clothing**	**$2,117,613.52**	**1.93 %**	**1.93 %**
⊞ **Components**	**$11,799,074.18**	**10.75 %**	**10.75 %**
Total	**$109,809,274.00**	**100.00 %**	**100.00 %**

```
OF_ All Categiry Sales (CM) = CALCULATE([Total Sales Amount (CM)],REMOVEFILTERS('Product'[Category]))

OF_ % Category Sales (CM) = DIVIDE([Total Sales Amount (CM)],[OF_ All Categiry Sales (CM)],0)

OF_ All Subcategory SAles (CM) =
CALCULATE(
  [Total Sales Amount (CM)],
  REMOVEFILTERS('Product'[Subcategory])

OF_ % SubCategiry Sales (CM) = DIVIDE([Total Sales Amount (CM)],[OF_ All Subcategory SAles (CM)],0)

OF_ Product Category ISINSCOPE (CM) = ISINSCOPE('Product'[Category])

OF_ Product SubCategory ISINSCOPE(CM) =ISINSCOPE('Product'[Subcategory])

OF_ % Product Sales Revised (CM) =
IF(
  [OF_ Product SubCategory ISINSCOPE(CM)],
  [OF_ % SubCategiry Sales (CM)],
  [OF_ % Category Sales (CM)]
)
```

Category	Total Sales Amount (CM)	A_ ALL Sales Amount (CM)	OF_ All Categiry Sales (CM)	OF_ % Category Sales (CM)	OF_ % SubCategiry Sales (CM)	OF_ Product Category ISINSCOPE (CM)	OF_ Product SubCategory ISINSCOPE(CM)	OF_ % Product Sales Revised (CM)
Accessories	**$1,272,059.71**	**$1,272,059.71**	**$109,809,274.00**	**1.16%**	**100.00%**	**True**	**False**	**1.16%**
Bike Racks	$237,096.21	$237,096.21	$237,096.21	100.00%	18.64%	True	True	18.64%
Bike Stands	$39,591.00	$39,591.00	$39,591.00	100.00%	3.11%	True	True	3.11%
Bottles and Cages	$64,274.95	$64,274.95	$64,274.95	100.00%	5.05%	True	True	5.05%
Cleaners	$18,407.03	$18,407.03	$18,407.03	100.00%	1.45%	True	True	1.45%
Fenders	$46,619.58	$46,619.58	$46,619.58	100.00%	3.66%	True	True	3.66%
Helmets	$484,049.97	$484,049.97	$484,049.97	100.00%	38.05%	True	True	38.05%
Hydration Packs	$105,826.62	$105,826.62	$105,826.62	100.00%	8.32%	True	True	8.32%
Lights						True	True	
Locks	$16,225.22	$16,225.22	$16,225.22	100.00%	1.28%	True	True	1.28%
Panniers						True	True	
Pumps	$13,514.71	$13,514.71	$13,514.71	100.00%	1.06%	True	True	1.06%
Tires and Tubes	$246,454.42	$246,454.42	$246,454.42	100.00%	19.37%	True	True	19.37%
Bikes	**$94,620,526.59**	**$94,620,526.59**	**$109,809,274.00**	**86.17%**	**100.00%**	**True**	**False**	**86.17%**
Clothing	**$2,117,613.52**	**$2,117,613.52**	**$109,809,274.00**	**1.93%**	**100.00%**	**True**	**False**	**1.93%**
Components	**$11,799,074.18**	**$11,799,074.18**	**$109,809,274.00**	**10.75%**	**100.00%**	**True**	**False**	**10.75%**
Total	**$109,809,274.00**	**$109,809,274.00**	**$109,809,274.00**	**100.00%**	**100.00%**	**False**	**False**	**100.00%**

OF_ Using ISINSCOPE with Variables(CM) =
VAR AllCategorySales =
CALCULATE([Total Sales Amount (CM)],REMOVEFILTERS('Product'[Category]))
VAR AllSubCategorySales =
CALCULATE([Total Sales Amount (CM)], REMOVEFILTERS('Product'[Subcategory]))
VAR PctCatSales =
DIVIDE([Total Sales Amount (CM)],AllCategorySales, 0)
VAR PctSubCatSales =
DIVIDE([Total Sales Amount (CM)],AllSubCategorySales,0)
RETURN
IF(ISINSCOPE('Product'[Category]),PctSubCatSales,PctCatSales)

```
OF_ Using ISINSCOPE with Variables(CM) =
VAR AllCategorySales =
    CALCULATE([Total Sales Amount (CM)],REMOVEFILTERS('Product'[Category]))
VAR AllSubCategorySales =
    CALCULATE([Total Sales Amount (CM)], REMOVEFILTERS('Product'[Subcategory]))
VAR PctCatSales =
    DIVIDE([Total Sales Amount (CM)],AllCategorySales, 0)
VAR PctSubCatSales =
    DIVIDE([Total Sales Amount (CM)],AllSubCategorySales,0)
RETURN
    IF(ISINSCOPE('Product'[Category]),PctSubCatSales,PctCatSales)
```

Category	Total Sales Amount (CM)	OF_ Using ISINSCOPE with Variables(CM)
Accessories	**$1,272,059.71**	**100.00%**
Bike Racks	$237,096.21	18.64%
Bike Stands	$39,591.00	3.11%
Bottles and Cages	$64,274.95	5.05%
Cleaners	$18,407.03	1.45%
Fenders	$46,619.58	3.66%
Helmets	$484,049.97	38.05%
Hydration Packs	$105,826.62	8.32%
Locks	$16,225.22	1.28%
Pumps	$13,514.71	1.06%
Tires and Tubes	$246,454.42	19.37%
Bikes	**$94,620,526.59**	**100.00%**
Mountain Bikes	$36,445,444.52	38.52%
Road Bikes	$43,878,791.78	46.37%
Touring Bikes	$14,296,290.29	15.11%
Clothing	**$2,117,613.52**	**100.00%**
Components	**$11,799,074.18**	**100.00%**
Total	**$109,809,274.00**	**100.00%**

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Parameter Table & Selected Value
Number pf Product (CT) =
DATATABLE(

```
  "Num Products", INTEGER,
  {
    {2},
    {3},
    {4},
    {5},
    {10},
    {15},
    {20}
  }
)
```

```
Slider_Number of Customer Buying Multiple Products (CM) =
VAR NumberOfProductSelected =
  SELECTEDVALUE('Number of Product (CT)'[Num Products],2)
VAR CustomersWithSelectedProducts =
FILTER(
  Customer,
  AND(
    Customer[CustomerKey] <> -1,
    CALCULATE(DISTINCTCOUNT(Sales[ProductKey] >= NumberOfProductSelected)
  )
)
VAR Result =
  COUNTROWS(CustomersWithSelectedProducts)
RETURN
  Result
```

```
Slider_Number of Customer Buying Multiple Products (CM) =
VAR NumberOfProductSelected =
    SELECTEDVALUE('Number of Product (CT)'[Num Products],2)
VAR CustomersWithSelectedProducts =
FILTER(
    Customer,
    AND(
        Customer[CustomerKey] <> -1,
        CALCULATE(DISTINCTCOUNT(Sales[ProductKey])) >= NumberOfProductSelected)
    )
)
VAR Result =
    COUNTROWS(CustomersWithSelectedProducts)
RETURN
    Result
```

Country	Slider_Number of Customer Buying Multiple Products (CM)
Australia	1064
Canada	510
France	255
Germany	285
United Kingdom	498
United States	743
Total	3355

Num Products

- ○ 2
- ○ 3
- ○ 4
- ● 5
- ○ 10
- ○ 15
- ○ 20

Customer Ranking (CC) = RANKX (Customer, [Total Sales Amount (CM)])

```
1 Customer Ranking (CC) =
2 RANKX (Customer, [Total Sales Amount (CM)])
```

hased CAL (CC)	Last Order Date (CC)	No of Sales (CC)	Minimum Sales Per Customer (CC)	Maximum Sales Per Customer(CC)	Customer Ranking (CC)
1	6/11/2020	1	4.99	4.99	17917
1	10/17/2019	1	4.99	4.99	17917
1	1/19/2020	1	4.99	4.99	17917
1	8/18/2019	1	4.99	4.99	17917
1	11/11/2019	1	4.99	4.99	17917
2	10/20/2019	2	2.29	4.99	17534
2	10/1/2019	2	2.29	4.99	17534
2	4/27/2020	2	2.29	4.99	17534
2	7/2/2019	2	2.29	4.99	17534
2	12/23/2019	2	2.29	4.99	17534
2	10/3/2019	2	2.29	4.99	17534
2	3/11/2020	2	2.29	4.99	17534
1	3/31/2020	1	9.99	9.99	17436
1	11/5/2019	1	9.99	9.99	17436
2	4/15/2020	2	4.99	8.99	17208
2	7/8/2019	2	4.99	8.99	17208
2	11/22/2019	2	4.99	8.99	17208
2	6/5/2020	2	4.99	9.99	17068
2	10/20/2019	2	4.99	9.99	17068

Customer	Total Sales Amount (CM)	Sum of Customer Ranking (CC)
[Not Applicable]	$80,450,596.11	1
Nichole Nara	$13,295.38	2
Kaitlyn Henderson	$13,294.27	3
Margaret He	$13,269.27	4
Randall Dominguez	$13,265.99	5
Adriana Gonzalez	$13,242.70	6
Rosa Hu	$13,215.65	7
Brandi Gill	$13,195.64	8
Brad She	$13,173.19	9
Francisco Sara	$13,164.64	10
Maurice Shan	$12,909.67	11
Janet Munoz	$12,489.17	12
Lisa Cai	$11,469.19	13
Lacey Zheng	$11,248.46	14
Larry Munoz	$11,068.01	16
Larry Vazquez	$10,899.62	17
Kate Anand	$10,872.06	18
Lawrence Alonso	$10,836.90	19
Terrance Rodriguez	$10,829.22	20
Aaron Wright	$10,813.63	21
Clarence Gao	$10,799.52	22
Bonnie Nath	$10,793.27	23
Andres Nara	$10,789.53	24
Ethan Bryant	$10,778.61	25
Ricky Vazquez	$10,580.35	26
Latasha Rubio	$10,575.33	27
Clarence Anand	$10,566.38	28

```
M Customer Ranking (CC) =
IF(
  Customer[CustomerKey]=-1,
  BLANK(),
  RANKX (Customer, [Total Sales Amount (CM)])
```

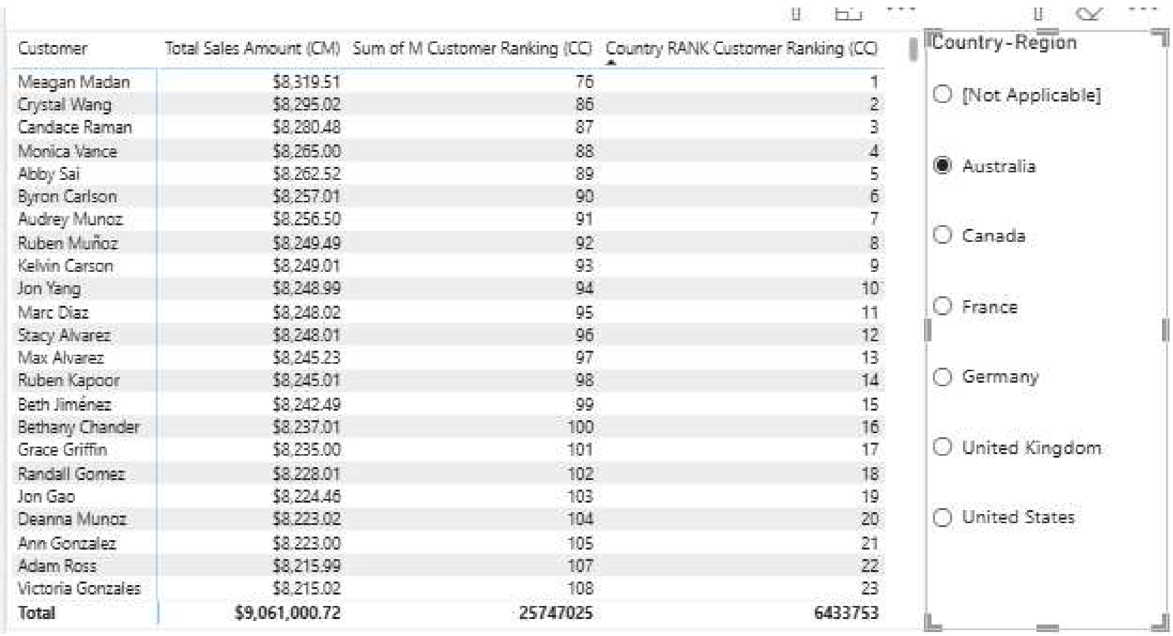
Customer	Total Sales Amount (CM)	Sum of Customer Ranking (CC)	Sum of M Customer Ranking (CC)
[Not Applicable]	$80,450,596.11	1	
Nichole Nara	$13,295.38	2	2
Kaitlyn Henderson	$13,294.27	3	3
Margaret He	$13,269.27	4	4
Randall Dominguez	$13,265.99	5	5
Adriana Gonzalez	$13,242.70	6	6
Rosa Hu	$13,215.65	7	7
Brandi Gill	$13,195.64	8	8
Brad She	$13,173.19	9	9
Francisco Sara	$13,164.64	10	10
Maurice Shan	$12,909.67	11	11
Janet Munoz	$12,489.17	12	12
Lisa Cai	$11,469.19	13	13
Lacey Zheng	$11,248.46	14	14
Larry Munoz	$11,068.01	16	16
Larry Vazquez	$10,899.62	17	17
Kate Anand	$10,872.06	18	18
Lawrence Alonso	$10,836.90	19	19
Terrance Rodriguez	$10,829.22	20	20
Aaron Wright	$10,813.63	21	21
Clarence Gao	$10,799.52	22	22
Bonnie Nath	$10,793.27	23	23
Andres Nara	$10,789.53	24	24
Ethan Bryant	$10,778.61	25	25
Ricky Vazquez	$10,580.35	26	26
Latasha Rubio	$10,575.33	27	27
Clarence Anand	$10,566.38	28	28
Total	**$109,809,274.00**	**170248797**	**170248796**

```
M Customer Ranking (CC) =
IF(
   Customer[CustomerKey]=-1,
   BLANK(),
   RANKX (Customer, [Total Sales Amount (CM)])-1
   )
```

Customer	Total Sales Amount (CM)	Sum of Customer Ranking (CC)	Sum of M Customer Ranking (CC)
[Not Applicable]	$80,450,596.11	1	
Nichole Nara	$13,295.38	2	1
Kaitlyn Henderson	$13,294.27	3	2
Margaret He	$13,269.27	4	3
Randall Dominguez	$13,265.99	5	4
Adriana Gonzalez	$13,242.70	6	5
Rosa Hu	$13,215.65	7	6
Brandi Gill	$13,195.64	8	7
Brad She	$13,173.19	9	8
Francisco Sara	$13,164.64	10	9
Maurice Shan	$12,909.67	11	10
Janet Munoz	$12,489.17	12	11
Lisa Cai	$11,469.19	13	12
Lacey Zheng	$11,248.46	14	13
Larry Munoz	$11,068.01	16	15
Larry Vazquez	$10,899.62	17	16
Kate Anand	$10,872.06	18	17
Lawrence Alonso	$10,836.90	19	18
Terrance Rodriguez	$10,829.22	20	19
Aaron Wright	$10,813.63	21	20
Clarence Gao	$10,799.52	22	21
Bonnie Nath	$10,793.27	23	22
Andres Nara	$10,789.53	24	23
Ethan Bryant	$10,778.61	25	24
Ricky Vazquez	$10,580.35	26	25
Latasha Rubio	$10,575.33	27	26
Clarence Anand	$10,566.38	28	27

Dynamic Ranking

(Dy Ranking)Country Wise Customer Ranking (CC) =
RANKX (
   ALLEXCEPT(Customer,Customer[Country-Region]),
   [Total Sales Amount (CM)]
)



Customer	Total Sales Amount (CM)	Sum of M Customer Ranking (CC)	Country RANK Customer Ranking (CC)
Meagan Madan	$8,319.51	76	1
Crystal Wang	$8,295.02	86	2
Candace Raman	$8,280.48	87	3
Monica Vance	$8,265.00	88	4
Abby Sai	$8,262.52	89	5
Byron Carlson	$8,257.01	90	6
Audrey Munoz	$8,256.50	91	7
Ruben Muñoz	$8,249.49	92	8
Kelvin Carson	$8,249.01	93	9
Jon Yang	$8,248.99	94	10
Marc Diaz	$8,248.02	95	11
Stacy Alvarez	$8,248.01	96	12
Max Alvarez	$8,245.23	97	13
Ruben Kapoor	$8,245.01	98	14
Beth Jiménez	$8,242.49	99	15
Bethany Chander	$8,237.01	100	16
Grace Griffin	$8,235.00	101	17
Randall Gomez	$8,228.01	102	18
Jon Gao	$8,224.46	103	19
Deanna Munoz	$8,223.02	104	20
Ann Gonzalez	$8,223.00	105	21
Adam Ross	$8,215.99	107	22
Victoria Gonzales	$8,215.02	108	23
Total	**$9,061,000.72**	**25747025**	**6433753**

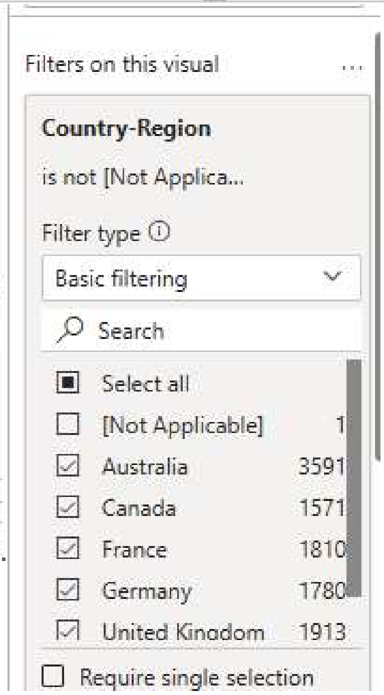
RANKX & ISINSCOPE

Geographical Rank (CM) =
IF(
   ISINSCOPE(Customer[State-Province]),
   RANKX(ALLSELECTED(Customer[State-Province]),[Total Sales Amount (CM)]),
   IF(
      ISINSCOPE(Customer[Country-Region]),
      RANKX(ALLSELECTED(Customer[Country-Region]),[Total Sales Amount (CM)])
)
)

```
Geographical Rand (CM) =
IF(
    ISINSCOPE(Customer[State-Province]),
    RANKX(ALLSELECTED(Customer[State-Province]),[Total Sales Amount (CM)]),
    IF(
        ISINSCOPE(Customer[Country-Region]),
        RANKX(ALLSELECTED(Customer[Country-Region]),[Total Sales Amount (CM)])
    )
)
```

Country-Region	Geographical Rand (CM)
[Not Applicable]	1
Australia	3
New South Wales	1
Queensland	3
South Australia	4
Tasmania	5
Victoria	2
Canada	7
France	6
Germany	5
United Kingdom	4
United States	2
Total	

Country Region	Geographical Rand (CM)
Australia	2
New South Wales	1
Queensland	3
South Australia	4
Tasmania	5
Victoria	2
Canada	6
France	5
Germany	4
United Kingdom	3
United States	1
Total	



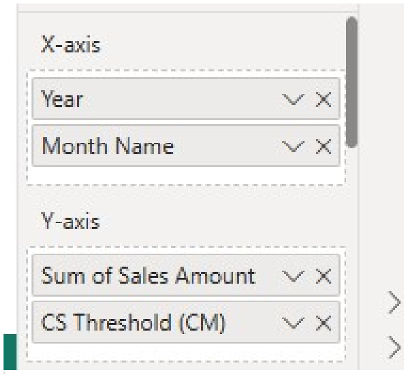
Counting spikes
Setting the threshold:

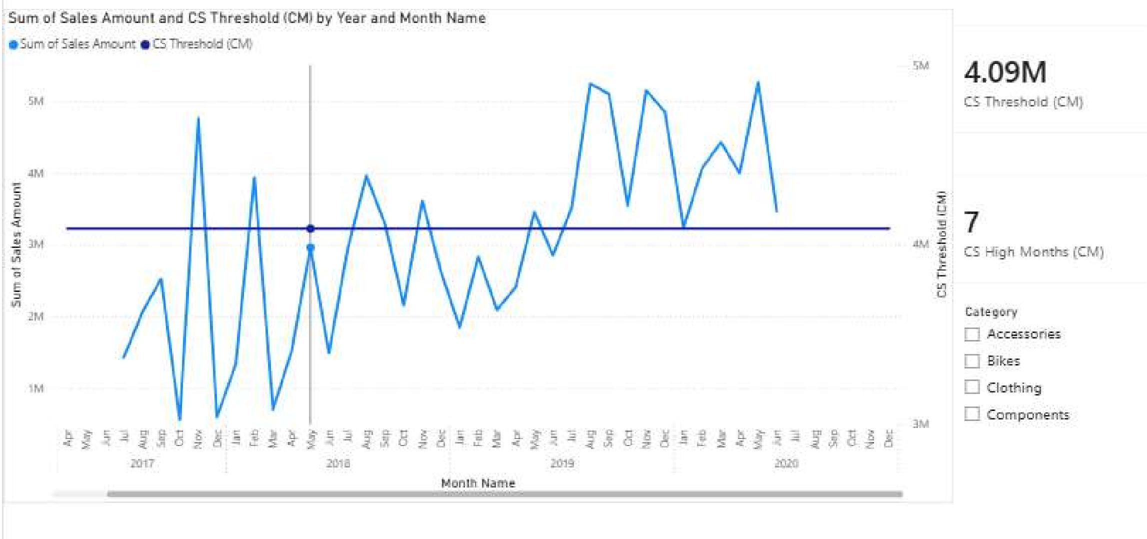
```
CS Threshold (CM) =
CALCULATE(
  VAR MinSales =
    MINX(ALL('Date'[Full Month]),[Total Sales Amount (CM)])
  VAR MaxSales =
    MAXX(ALL('Date'[Full Month]),[Total Sales Amount (CM)])
  VAR Boundary = MinSales +(MaxSales - MinSales) * 0.75
  RETURN
    Boundary,
    ALL('Date')
)
```

Finding the High Months

```
CS High Months (CM) =
```

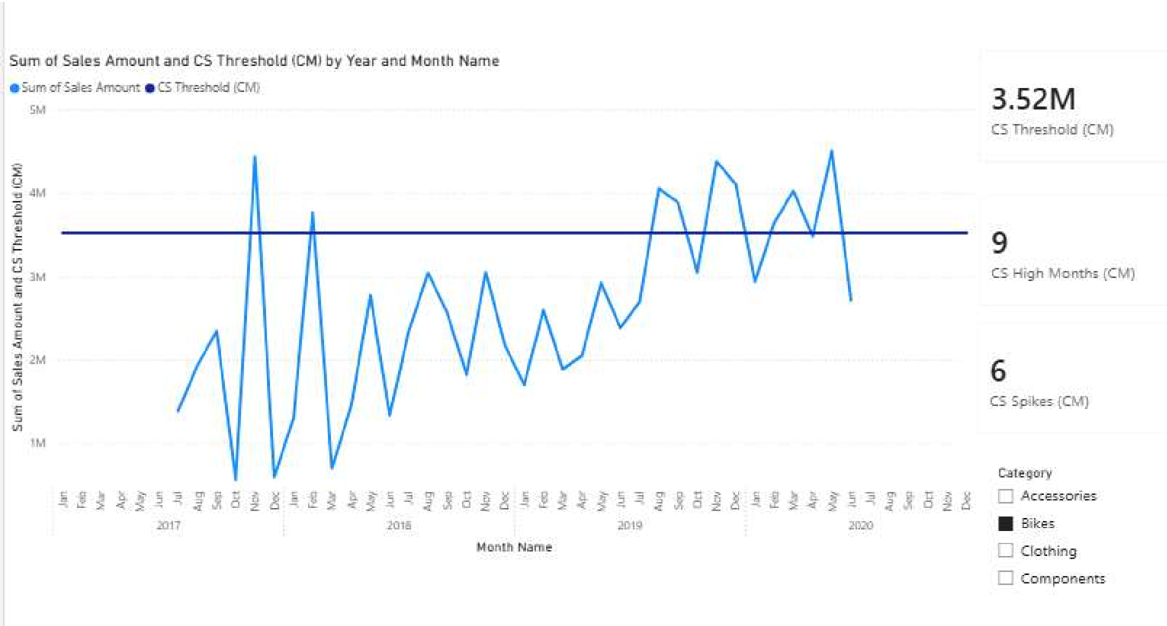
```
VAR Threshold = [CS Threshold (CM)]
VAR MonthsOverThreshold =
  FILTER(ALL('Date'[Full Month]),[Total Sales Amount (CM)]> Threshold)
VAR Result =
  COUNTROWS(MonthsOverThreshold)
RETURN
  Result
```





Using PreviousMonth()

```
CS Spikes (CM) =
VAR Threshold = [CS Threshold (CM)]
VAR MaxSpikes =
  FILTER(
    VALUES('Date'[Full Month]),
    VAR SalesCurrentMonth = [Total Sales Amount (CM)]
    VAR SalesPreviousMonth =
      CALCULATE([Total Sales Amount (CM)],PREVIOUSMONTH('Date'[Date]))
    VAR EnteringThreshold = SalesCurrentMonth >=Threshold
      && SalesPreviousMonth < Threshold
    RETURN
      EnteringThreshold
  )
VAR Result = COUNTROWS(MaxSpikes)
RETURN
  Result
```



---- Hierarchies---

DAX Mastery – EMP-Sales

Full Path (CC) =
PATH(Employees[Employee ID],Employees[Reports to])

Using PATHITEM()

PATHITEM() searches the Path column and returns the key present at the Nth position of the path.

Using LOOKUPVALUE()

The function is used to the return the name of the Employee whose Employee ID matches with the searched value by PATHITEM()

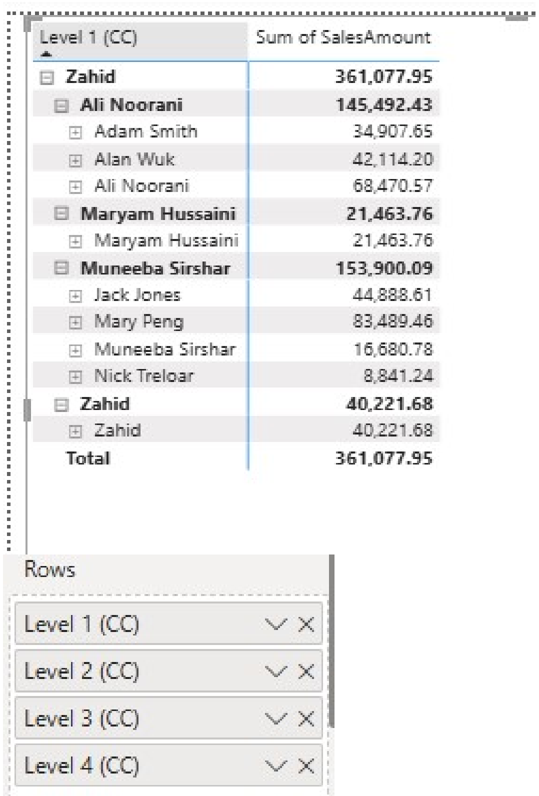
Level Column

```
Level 1 (CC)=
LOOKUPVALUE(
  Employees[Names],
  Employees[Employee ID], PATHITEM(Employees[Full Path (CC)],1,INTEGER)
)
```

```
Level 2 (CC)=
IF(
  PATHLENGTH(Employees[Full Path (CC)]) >=2,
  LOOKUPVALUE(
    Employees[Names],
    Employees[Employee ID],PATHITEM(Employees[Full Path (CC)],2,INTEGER)
  ),
  Employees[Level 1 (CC)]
)
```

```
Level 3 (CC)=
IF(
  PATHLENGTH(Employees[Full Path (CC)]) >= 3,
  LOOKUPVALUE(
    Employees[Names],
    Employees[Employee ID],PATHITEM(Employees[Full Path (CC)],3,INTEGER)
  ),
  Employees[Level 2 (CC)]
)
```

```
Level 4 (CC)=
IF(
  PATHLENGTH(Employees[Full Path (CC)]) >= 4,
  LOOKUPVALUE(
    Employees[Names],
    Employees[Employee ID],PATHITEM(Employees[Full Path (CC)],4,INTEGER)
  ),
  Employees[Level 3 (CC)]
)
```



Level 1 (CC)	Sum of SalesAmount
Zahid	**361,077.95**
Ali Noorani	**145,492.43**
Adam Smith	34,907.65
Alan Wuk	42,114.20
Ali Noorani	68,470.57
Maryam Hussaini	**21,463.76**
Maryam Hussaini	21,463.76
Muneeba Sirshar	**153,900.09**
Jack Jones	44,888.61
Mary Peng	83,489.46
Muneeba Sirshar	16,680.78
Nick Treloar	8,841.24
Zahid	**40,221.68**
Zahid	40,221.68
Total	**361,077.95**

Finding Maximum Depths

Node Depth (CC) =PATHLENGTH(Employees[Full Path (CC)])

Determining Browsing Levels (Calculated Measures)

Maxmimum Node Depth (CM) =MAX(Employees[Node Depth (CC)])

Level 1 (CC)	Sum of SalesAmount	Maxmimum Node Depth (CM)
Zahid	**361,077.95**	**4**
Ali Noorani	**145,492.43**	**3**
Adam Smith	34,907.65	3
Alan Wuk	42,114.20	3
Ali Noorani	68,470.57	2
Maryam Hussaini	**21,463.76**	**2**
Maryam Hussaini	21,463.76	2
Muneeba Sirshar	**153,900.09**	**4**
Jack Jones	44,888.61	3
Mary Peng	83,489.46	4
Muneeba Sirshar	16,680.78	2
Nick Treloar	8,841.24	3
Zahid	**40,221.68**	**1**
Zahid	40,221.68	1
Total	**361,077.95**	**4**

Browsing Depth (CM) =
ISINSCOPE(Employees[Level 1 (CC)])
+ISINSCOPE(Employees[Level 2 (CC)])
+ISINSCOPE(Employees[Level 3 (CC)])
+ISINSCOPE(Employees[Level 4 (CC)])

```
Browsing Depth (CM) =
ISINSCOPE(Employees[Level 1 (CC)])
+ISINSCOPE(Employees[Level 2 (CC)])
+ISINSCOPE(Employees[Level 3 (CC)])
+ISINSCOPE(Employees[Level 4 (CC)])
```

Level 1 (CC)	Sum of SalesAmount	Maxmimum Node Depth (CM)	Browsing Depth (CM)
Zahid	**361,077.95**	**4**	**1**
Ali Noorani	**145,492.43**	**3**	**2**
Adam Smith	34,907.65	3	3
Alan Wuk	42,114.20	3	3
Ali Noorani	68,470.57	2	3
Maryam Hussaini	**21,463.76**	**2**	**2**
Maryam Hussaini	21,463.76	2	3
Muneeba Sirshar	**153,900.09**	**4**	**2**
Jack Jones	44,888.61	3	3
Mary Peng	83,489.46	4	3
Muneeba Sirshar	16,680.78	2	3
Nick Treloar	8,841.24	3	3
Zahid	**40,221.68**	**1**	**2**
Zahid	40,221.68	1	3
Total	**361,077.95**	**4**	**0**

Analyzing Sales in Hierarchies

Sales at Level (CM) =
IF(

```
    MAX(Employees[Node Depth (CC)]) < [Browsing Depth (CM)],
    BLANK(),
    SUM(Sales[SalesAmount])
)
```

Level 1 (CC)	Sum of SalesAmount	Maxmimum Node Depth (CM)	Browsing Depth (CM)	Sales at Level (CM)
Zahid	**361,077.95**	**4**	**1**	**$361,077.95**
Zahid	**40,221.68**	**1**	**2**	
Muneeba Sirshar	**153,900.09**	**4**	**2**	**$153,900.09**
Nick Treloar	8,841.24	3	3	$8,841.24
Muneeba Sirshar	16,680.78	2	3	
Mary Peng	83,489.46	4	3	$83,489.46
Jack Jones	44,888.61	3	3	$44,888.61
Maryam Hussaini	**21,463.76**	**2**	**2**	**$21,463.76**
Ali Noorani	**145,492.43**	**3**	**2**	**$145,492.43**
Total	**361,077.95**	**4**	**0**	**$361,077.95**

Removed Maxximum Node Depth, & Browsing Depth to get the Expected Visualization

Level 1 (CC)	Sales at Level (CM)
Zahid	**$361,077.95**
Muneeba Sirshar	**$153,900.09**
Nick Treloar	$8,841.24
Mary Peng	$83,489.46
Jack Jones	$44,888.61
Maryam Hussaini	**$21,463.76**
Ali Noorani	**$145,492.43**
Alan Wuk	$42,114.20
Adam Smith	$34,907.65
Total	**$361,077.95**